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Four Decades of Implicit Monte Carlo

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Abstract

In 1971, Fleck and Cummings derived a system of equations to enable robust Monte Carlo simulations of time-dependent, thermal radiative transfer problems. Denoted the “Implicit Monte Carlo” (IMC) equations, their solution remains the de facto standard of high-fidelity radiative transfer simulations. Over the course of forty-four years, their numerical properties have become better understood, and accuracy enhancements, novel acceleration methods, and variance reduction techniques have been suggested. In this review, we re-derive the IMC equations – explicitly highlighting assumptions as they are made – and outfit the equations with a Monte Carlo interpretation. We put the IMC equations in context with other approximate forms of the radiative transfer equations and present a new demonstration of their equivalence to another well-used linearization solved with deterministic transport methods for frequency-independent problems. We discuss physical and numerical limitations of the IMC equations for asymptotically small time steps, stability characteristics and the potential of maximum principle violations for large time steps, and solution behaviors in an asymptotically thick diffusive limit. We provide a new stability analysis for opacities with general monomial dependence on temperature. We consider spatial accuracy limitations of the IMC equations and discussion acceleration and variance reduction techniques.

Keywords: Implicit Monte Carlo, thermal radiative transfer

1. Introduction

More than forty years have passed since the “Implicit Monte Carlo” (IMC) equations were proposed by Fleck and Cummings as the basis of a Monte Carlo solution methodology for problems requiring time-dependent, nonlinear, thermal radiative transfer (TRT) physics (Fleck & Cummings, 1971). These problems necessitate the solution of the radiation *specific intensity*, I , that is nonlinearly coupled to the *material internal energy* $U_m(\mathbf{x}, t)$, which, for our purposes, is well-described by a *temperature* field $T(\mathbf{x}, t)$. Examples of such problems occur in astrophysical phenomena, inertial confinement fusion problems, oxy-coal combustion, scramjets, and other high-temperature heat transfer with participating media (Cleveland, 2011; Drake et al., 2011; Heymann & Siebenmorgen, 2012; Howell, 1998; Iliev et al., 2006; Irvine, 2013; Kasen et al., 2006; Petkova & Springel, 2009; Zimmerman, 1997). The IMC equations have even been extended to neutrino transport (Abdikamalov et al., 2012; Wollaeger et al., 2013). The hallmark of the IMC equations is that they approximate the rapid, dynamic timescale of photon absorption and re-emission via “effective” scattering events, so that as time step sizes are increased, so does the prevalence of effective scattering. For problem regions that are sufficiently thick and whose intensities are nearly black-body, the timescale describing $O(1)$ changes in temperature evolution is asymptotically slower by a factor of ϵ^2 (as $\epsilon \rightarrow 0$) as compared with the absorption and re-emission timescale (Larsen et al., 1983), so this effective scattering treatment can enable robust, accurate solutions that require significantly fewer time steps than an explicit time discretization. It is likely that this feature has supported the longevity of the IMC methodology. However, throughout the last several decades, certain solution and algorithmic deficiencies have emerged and received due treatment; an

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overview of these from the vantage points of the time and spatial discretizations, algorithmic improvements, and variance reduction follows.

The IMC equations are *not* the end-equations that one would derive by performing a truly implicit time discretization of the TRT equations. More precisely, they are semi-implicit, so “IMC” is a misnomer. This semi-implicit treatment recasts the nonlinear coupling of specific intensity and temperature into a triangular system for I and T over a specified time step; that is, the end-of-time-step specific intensity may be solved via standard transport techniques (such as linear Monte Carlo or deterministic discrete ordinates methods), after which the end-of-time-step temperature may be updated using the just-computed transport solution. The advantage of this approach is that it circumvents the need to perform a coupled, nonlinear solution of the TRT equations without giving up unconditional stability. This nominally introduces an $O(\Delta t)$ error as the time step size Δt is asymptotically reduced to zero (Cheatham, 2010; Martin & Brown, 2001a). Some more subtle consequences are that the IMC equations admit damped oscillatory solutions in time, their temperature solutions can violate a maximum principle (internal temperatures can exceed boundary temperatures in the absence of sources for sufficiently large time steps), and they do not preserve an asymptotic equilibrium diffusion limit (Densmore & Larsen, 2004; Larsen et al., 2013; Larsen & Mercier, 1987; Martin & Brown, 2001a; Wollaber & Larsen, 2011; Wollaber et al., 2013). These issues can be ameliorated by reducing the time step size, although it is difficult to determine a useful, a priori, upper-bound on Δt .

In addition to temporal discretization error, any practical solution of the IMC equations contains *spatial* truncation errors that arise due to the discrete, spatial representation of the time-evolving temperature field. We note that a Monte Carlo solution of the IMC equations does not necessitate a spatial grid for the specific intensity; the interplay between the continuous intensity representation and the closely coupled, discrete temperature field complicates the overall spatial truncation analysis except in special limits (Densmore, 2011, 2012, 2013). These spatial truncation errors can give rise to so-called “teleportation” effects, in which Monte Carlo particles are absorbed and non-physically re-emitted ahead of a wavefront, artificially increasing the wave speed (Cheatham, 2010; Densmore, 2011; Irvine, 2013; Long, 2012; McKinley et al., 2003). It was empirically found that (at least) linear shape functions are necessary to mitigate this effect (Fleck & Canfield, 1984), and later analyses conclusively determined that a histogram temperature field is inadequate (Densmore, 2011). The magnitude of teleportation errors can be reduced by decreasing the spatial grid size and/or increasing the time step size; however, if the problem is near a regime in which maximum principle violations might occur, then this prescription can, in turn, exacerbate those undesirable features (Wollaber et al., 2013). Consequently, the IMC practitioner should not blindly adjust spatial or temporal grid sizes to successfully perform a mesh-resolution study, even though the IMC grid parameters are not Courant-limited in the traditional sense.

When the IMC equations are solved using a Monte Carlo methodology, a predominance of effective scattering in optically thick zones can dramatically slow the time-to-solution, as a profusion of short particle flights and scattering events must be simulated for each particle history (in deterministic algorithms, these problems can effectively be accelerated using diffusion synthetic acceleration or related methodologies). Relatively early on, it was recognized that times-to-solution could be reduced at little to no cost in accuracy by employing an on-the-fly diffusion approximation in spheres inscribed within those zones: the so-called “random walk” method (Cooper, 2000; Fleck & Canfield, 1984; Giorla & Sentis, 1987). More recently, several “discrete diffusion Monte Carlo” approaches have emerged that entirely replace transport zones with diffusive zones (Cleveland & Gentile, 2014; Cleveland et al., 2010; Densmore et al., 2012; Gentile, 2001). These techniques have been shown to accelerate calculations by orders of magnitude in some cases.

Finally, the Monte Carlo solutions of the IMC equations are subject to statistical variance. For a sample mean (tally) being computed over a given time step, the magnitude of the variance in any tallied quantity can be systematically decreased by a factor of $1/\sqrt{N}$, where N is the number of Monte Carlo particles, due to arguments stemming from the central limit theorem. Often, this is not sufficient to produce solutions of reasonable quality in a given amount of compute-time, particularly for tallies that require the sampling of low-probability events. To complicate this, IMC problems are typically semi-*global* in their spatial domains; that is, it is usually important to obtain reasonable-quality solutions throughout a major fraction of the problem geometry (this is in contrast to *local* Monte Carlo problems in which a response is sought in a

small fraction of the problem phase space). Reducing statistical noise in multi physics problems can be crucial to obtaining correct solution behavior, as the radiation noise can seed instabilities in the other physics (Gentile, 2009). Because of this, some standard variance reduction techniques are usually employed, including “implicit capture”, particle combing, and modifications to particle sourcing (Booth, 1996; Fleck & Cummings, 1971; Gray et al., 1998; Kuilman et al., 2012; Urbatsch, 2012). We will also address more advanced techniques such as weight windows (Cooper & Larsen, 2001; Gentile, 2009; Wollaber & Larsen, 2009a; Zimmerman, 1997).

The remainder of this review is organized as follows. In Section 2, we derive the IMC equations from the thermal radiative transfer equations and outfit them with a Monte Carlo interpretation. Throughout this section, we will discuss the IMC equations’ robustness and contrast the approximations made to obtain the equations with competing methodologies. In Section 4, we will consider their temporal accuracy, both for short and long time steps. In Section 5, we will consider spatial grid effects on solution accuracy. Section 6 summarizes the state of acceleration methods and algorithms for computing Monte Carlo solutions of the IMC equations as well as energy-conserving variance reduction techniques.

2. Background

To put the IMC equations and solution methodology in context, we will derive them from the thermal radiative transfer (TRT) equations. First, we will define the TRT system under consideration, recast it into a useful form for further analysis, and discuss some of its solution characteristics. We then proceed with the IMC derivation, highlighting where approximations are made and putting them into context with competing methodologies. The section concludes with a brief discussion on energy conservation and positivity in the IMC equations.

We begin from a relatively simple form of the TRT equations that describe the coupling of radiation with matter. We shall assume that the problem is prescribed in a volume V with spatial coordinate $\mathbf{x} \in V$ whose outer surface is ∂V . The radiation intensity will be allowed to vary with respect to spatial coordinate $\mathbf{x}$, solid angle $\boldsymbol{\Omega} \in 4\pi$, and frequency $\nu \in [0, \infty)$ for times $t > 0$. We assume that the radiation is sufficiently energetic that wave effects such as diffraction, polarization, and refraction may be neglected, and that the radiation-matter interaction arises due to volumetric (not surface) effects. We further assume that the material emissivity is well-described by a black body at temperature T (this is sometimes referred to as the material being in “local thermodynamic equilibrium” (LTE) in this field, not to be confused with LTE implying full matter-radiation equilibrium as in other fields). We neglect thermal conduction, hydrodynamic motion, explicit ion-electron coupling, and relativistic effects in our presentation. Many of these physics can be fitted to the fundamental IMC methodology, but they would muddy its presentation. Finally, we do not formally include physical scattering in our derivation; reincorporating Thomson scattering is straightforward, and there are also techniques to implement Compton scattering (Carter & Forest, 1973; Castor, 2004; Densmore et al., 2010, 2009, 2007; Dolence et al., 2009; Evans & Densmore, 2007; Fleck, 1971; Hungerford et al., 2005; Pomraning, 1973). The equations that model this system are

$$\frac{1}{c} \frac{\partial I}{\partial t} + \boldsymbol{\Omega} \cdot \nabla I + \sigma I = \sigma B + \frac{Q_r}{4\pi}, \quad (1a)$$

$$c_v \frac{\partial T}{\partial t} = \int_0^\infty \int_{4\pi} \sigma (I - B) d\Omega' d\nu' + Q_m, \quad (1b)$$

for $\mathbf{x} \in V$, $\boldsymbol{\Omega} \in 4\pi$, $0 < \nu < \infty$, and $0 < t$. We also impose initial conditions

$$I(\mathbf{x}, \boldsymbol{\Omega}, \nu, 0) = I^i(\mathbf{x}, \boldsymbol{\Omega}, \nu), \quad (1c)$$

$$T(\mathbf{x}, 0) = T^i(\mathbf{x}), \quad (1d)$$

and boundary conditions

$$I(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = I^b(\mathbf{x}, \boldsymbol{\Omega}, \nu, t), \text{ for } \mathbf{x} \in \partial V \text{ and } \boldsymbol{\Omega} \cdot \mathbf{n} < 0. \quad (1e)$$

In these equations, we have defined

$$I = ch\nu n(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = \text{the specific intensity, GJ/cm}^2\text{-ns-keV-sr, where} \quad (2a)$$

$$n(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = \text{the mean number of photons per unit of phase space,} \quad (2b)$$

$$h = 4.135667516 \times 10^{-9} \text{ keV-ns} = \text{Planck's constant,} \quad (2c)$$

$$c = 29.9792458 \text{ cm/ns} = \text{speed of light,} \quad (2d)$$

$$T = T(x, t) = \text{the material temperature (keV),} \quad (2e)$$

$$B = B(\nu, T) = \text{Planck's function for radiation, GJ/cm}^2\text{-ns-keV-sr,} \\ = \frac{2T^3}{h^3 c^2} \frac{(\nu/T)^3}{e^{\nu/T} - 1}, \quad (2f)$$

$$Q_r = Q_r(\mathbf{x}, \nu, t) = \text{inhomogeneous photon source, GJ/cc-ns-keV,} \quad (2g)$$

$$Q_m = Q_m(\mathbf{x}, t) = \text{inhomogeneous material energy source, GJ/cc-ns,} \quad (2h)$$

$$c_v = c_v(\mathbf{x}, T) = \text{heat capacity of material, GJ/keV-cc} \quad (2i)$$

$$\sigma_a = \sigma_a(\mathbf{x}, \nu, T) = \text{the absorption opacity, cm}^{-1}, \\ = (\text{probability of absorption per unit distance}). \quad (2j)$$

The quantities I and T are unknown; all of the other quantities are prescribed. In addition, it is useful to define the *material energy density* U_m and the *equilibrium radiation energy density* U_r , which are given by the relationships

$$\frac{\partial U_m}{\partial T}(\mathbf{x}, t) = c_v(\mathbf{x}, T), \quad (3a)$$

$$U_r(\mathbf{x}, t) = aT(\mathbf{x}, t)^4, \quad (3b)$$

where

$$a = \text{radiation constant} = \frac{8\pi^5 k^4}{15h^3 c^3} = 0.01372 \frac{\text{GJ}}{\text{cm}^3 \text{ keV}^4}. \quad (3c)$$

Our choice of units (GJ for energy, ns for time, keV for temperature and frequency) is intended to simplify the expression of data in the high energy density physics regime (regimes with pressures greater than 100 MPa and/or temperatures on the order of one million K (Drake, 2006)). In particular, the Planck spectrum is written with both frequency ν and temperature T in units of keV by substituting $\nu \leftarrow h\hat{\nu}$ and $T \leftarrow k\hat{T}$, where the “hat” variables are in their traditional form of Hz and K, respectively, and k is Boltzmann’s constant. The factor of h^3 (not h^2) appears in the denominator of Eq. (2f) by performing this substitution in the differential element $d\nu$. The functional dependencies of the heat capacity c_v and absorption opacity σ , written as $(\mathbf{x}, T)$, are actually dependent upon the material density ρ and temperature T (or any two thermodynamic quantities).

2.1. Recasting the TRT Equations

Before proceeding with the derivation, it is useful to rewrite the TRT equations in an equivalent form that is more amenable to Monte Carlo interpretation and linearization and to formally demonstrate energy conservation. First, we define a frequency-normalized Planck spectrum $b(\nu, T)$, by using Eq. (2f),

$$b(\nu, T) \equiv \frac{B(\nu, T)}{\int_0^\infty B(\nu, T) d\nu}, \quad (4)$$

and, noting that,

$$\int_0^\infty \frac{(\nu/T)^3}{e^{\nu/T} - 1} d\nu = T \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx = T \int_0^\infty x^3 \sum_{n=1}^\infty e^{-nx} dx, \quad (5)$$

and, integrating by parts and taking the sum,

$$\int_0^\infty \frac{(\nu/T)^3}{e^{\nu/T} - 1} d\nu = T \sum_{n=1}^\infty \frac{6}{n^4} = T \frac{\pi^4}{15}. \quad (6)$$

Combining Eqs. (2f), (3c), and (6) (with k^4 used to convert units), the frequency-integrated Planck spectrum is

$$\int_0^\infty B(\nu, T) d\nu = \frac{2\pi^4 T^4}{15h^3 c^2} = \frac{caT^4}{4\pi}, \quad (7)$$

and the frequency-normalized Planck spectrum from Eq. (4) and (7) is

$$b(\nu, T) = \frac{15}{\pi^4} \frac{1}{T} \frac{(\nu/T)^3}{e^{\nu/T} - 1}, \quad (8)$$

which, by definition, satisfies $\int_0^\infty b(\nu, T) d\nu = 1$.

The *Planck opacity* may now be defined by using the normalized Planck spectrum as a spectral weighting function to evaluate a gray opacity,

$$\sigma_p \equiv \int_0^\infty \sigma(\nu, T) b(\nu, T) d\nu. \quad (9)$$

We also define an exact relationship between U_m and U_r via Eqs. (3),

$$\beta = \beta(\mathbf{x}, t) = \frac{\partial U_r}{\partial U_m} = \frac{dU_r/dT}{dU_m/dT} = \frac{4aT^3}{c_v(T)}, \quad (10)$$

Using the definitions in Eqs. (3), (4), (9), and (10), the TRT equations given as Eqs. (1) become

$$\frac{1}{c} \frac{\partial I}{\partial t} + \boldsymbol{\Omega} \cdot \nabla I + \sigma I = \sigma b \frac{cU_r}{4\pi} + \frac{Q_r}{4\pi}, \quad (11a)$$

$$\frac{\partial U_m}{\partial t} + \sigma_p c U_r = \int_0^\infty \int_{4\pi} \sigma I d\Omega' d\nu' + Q_m, \quad (11b)$$

and, equivalently,

$$\frac{1}{\beta} \frac{\partial U_r}{\partial t} + \sigma_p c U_r = \int_0^\infty \int_{4\pi} \sigma I d\Omega' d\nu' + Q_m. \quad (11c)$$

Although it may seem redundant to write Eqs. (11b) and (11c) in mathematically equivalent forms, approximations are typically made to them such that they lose their equivalency (in particular, Eq. (11c) now appears linear in T^4 , if only β and σ weren't temperature-dependent!). The TRT equations given by Eqs. (11) serve as our starting point for subsequent derivations. We next consider some characteristics of solutions of the TRT equations.

When the inhomogeneous sources Q_m and Q_r are zero, then solutions of Eqs. (11) satisfy a *maximum principle* (Andreev et al., 1983). Essentially, this implies that the interior solutions I and T are bounded above and below by the largest and smallest boundary conditions, respectively, assuming that the initial conditions are also so limited. In practice, this implies that if a numerically obtained interior temperature is found to exceed the boundary condition temperature, then it is not an admissible solution of the TRT equations.

Eqs. (11) admit the equilibrium (steady-state) solutions $I = B(\nu, T_0)$ and $T = T_0$ for any prescribed temperature T_0 if the inhomogeneous sources Q_m and Q_r are zero and the boundary conditions are chosen such that $I^b = B(\nu, T_0)$.

Eqs. (11) may also be manipulated to produce an energy conservation equation. We demonstrate this by defining the energy density and flux,

$$E \equiv \frac{1}{c} \int_0^\infty \int_{4\pi} I d\Omega d\nu, \quad (12a)$$

$$\mathbf{F} \equiv \int_0^\infty \int_{4\pi} \boldsymbol{\Omega} I d\Omega d\nu, \quad (12b)$$

by integrating Eq. (11a) over all solid angle and frequencies, and by summing the result with Eq. (11b),

$$\frac{\partial}{\partial t} (E + U_m) + \nabla \cdot \mathbf{F} = Q_m + \int_0^\infty Q_r d\nu. \quad (13)$$

This expression shows that the time-rate of change in the total energy content is given by streaming losses ($\nabla \cdot \mathbf{F}$) and contributions from internal sources; the matter-radiation energy exchange has cancelled out. This implies that it is possible to make consistent approximations to the absorption-emission physics without affecting energy conservation. Energy conservation holds for all times $t > 0$, but we may also demonstrate this by integrating over an arbitrary time step $t_n \leq t \leq t_{n+1}$, with $\Delta t = t_{n+1} - t_n$. Defining a time-averaging operator notation as

$$\overline{(\cdot)} \equiv \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} (\cdot) dt, \quad (14)$$

Eq. (13) is

$$\frac{\Delta E}{\Delta t} + \frac{\Delta U_m}{\Delta t} + \nabla \cdot \overline{\mathbf{F}} = \overline{Q_m} + \int_0^\infty \overline{Q_r} d\nu. \quad (15)$$

This equivalent form of energy conservation is a useful reference point for time-discretized methodologies.

2.2. The IMC Equations

We next make a series of approximations to Eqs. (11) in order to arrive at a linear transport equation coupled to a material energy update equation. The first approximation, common to most techniques, is to freeze temperature-dependent problem data at the beginning of a time step $t_n \leq t \leq t_{n+1}$:

$$\sigma_n \equiv \sigma(\mathbf{x}, \nu, T(t_n)) \approx \sigma_a(x, \nu, T(t)), \quad (16a)$$

$$b_n \equiv b(\nu, T(t_n)) \approx b(\nu, T(t)), \quad (16b)$$

$$\sigma_{p,n} \equiv \sigma(x, T(t_n)) \approx \sigma_p(x, T(t)), \quad (16c)$$

$$\beta_n \equiv \beta(\mathbf{x}, T(t_n)) \approx \beta(\mathbf{x}, T(t)). \quad (16d)$$

These approximations make Eqs. (11a) and (11c) a system of *linear*, first-order differential equations in U_r (T^4) and I ,

$$\frac{1}{c} \frac{\partial I}{\partial t} + \boldsymbol{\Omega} \cdot \nabla I + \sigma_n I = \sigma_n b_n \frac{c U_r}{4\pi} + \frac{Q_r}{4\pi}, \quad (17a)$$

$$\frac{\partial U_m}{\partial t} + \sigma_{p,n} c U_r = \int_0^\infty \int_{4\pi} \sigma_n I d\Omega' d\nu' + Q_m, \quad (17b)$$

and, *not* equivalently,

$$\frac{1}{\beta_n} \frac{\partial U_r}{\partial t} + \sigma_{p,n} c U_r = \int_0^\infty \int_{4\pi} \sigma_n I d\Omega' d\nu' + Q_m. \quad (17c)$$

The differences between Eqs. (11) and (17) may seem slight, but this approximation is arguably the most severe of those required to arrive at the IMC equations.

The next task is to approximate the unknown emission term U_r in Eq. (17a) by time-discretizing Eq. (17c) and eliminating its dependence on $I(t)$. Operating on Eqs. (17c) with the time-averaging operator defined in Eq. (14) produces the exact, time-integrated representation,

$$\frac{1}{\beta_n} \frac{U_{r,n+1} - U_{r,n}}{\Delta t} + \sigma_{p,n} c \bar{U}_r = \int_0^\infty \int_{4\pi} \sigma_n \bar{I} d\Omega' d\nu' + \bar{Q}_m. \quad (18)$$

A time-implicit approximation at this point would impose $\bar{U}_r \approx U_{r,n+1}$ and $\bar{I} \approx I_{n+1}$. Instead, we approximate $\bar{U}_r$ by introducing the parameter α and representing $\bar{U}_r$ as a linear combination of its beginning- and end-of-time step values,

$$\bar{U}_r \approx (1 - \alpha)U_{r,n} + \alpha U_{r,n+1}. \quad (19)$$

If $U_r(t)$ is continuous in time, then there exists an $\alpha \in [0, 1]$ such that Eq. (19) is exact. However, α is typically set to 1.0 to minimize temporal oscillations and must be 0.5 or greater in practice to ensure unconditional stability; this will be addressed in more detail in Section 4. Continuing, we substitute Eq. (19) into Eq. (18) and “solve” for $\bar{U}_r$,

$$\bar{U}_r = \frac{1}{1 + \alpha \sigma_{p,n} c \Delta t} U_{r,n} + \frac{\alpha \Delta t \beta_n}{1 + \alpha \sigma_{p,n} c \Delta t} \left(\int_0^\infty \int_{4\pi} \sigma_n \bar{I} d\Omega' d\nu' + \bar{Q}_m \right). \quad (20)$$

We next identify and define the *Fleck factor* f ,

$$f_n = \frac{1}{1 + \alpha \sigma_{p,n} c \Delta t}, \quad (21)$$

and rewrite Eq. (20) more compactly as

$$\bar{U}_r = f_n U_{r,n} + \frac{1}{c \sigma_{p,n}} \int_0^\infty \int_{4\pi} (1 - f_n) \sigma_n \bar{I} d\Omega' d\nu' + \frac{1}{c \sigma_{p,n}} (1 - f_n) \bar{Q}_m. \quad (22)$$

Eq. (22) provides the time-averaged U_r as a function of the time-averaged intensity I . However, the transfer equation Eq. (17a) requires the time-continuous $U_r(t)$. This leads us to the third approximation: the time-average quantities are replaced with their time-continuous counterparts,

$$U_r(t) \approx f_n U_{r,n} + \frac{1}{c \sigma_{p,n}} \int_0^\infty \int_{4\pi} (1 - f_n) \sigma_n I(t) d\Omega' d\nu' + \frac{1}{c \sigma_{p,n}} (1 - f_n) Q_m. \quad (23)$$

Eqs. (22) and (23) are consistent in that the time-average of Eq. (23) is Eq. (22), but Eq. (23) is clearly an approximation for arbitrary t (consider that $U_r(t_n) \neq U_{r,n}$ in Eq. (23)). Substituting Eq. (23) into Eqs. (17a) and Eqs. (17b) produces the IMC equations,

$$\frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I = \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \int_0^\infty \int_{4\pi} (1 - f_n) \sigma_n I d\Omega' d\nu' + f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{c U_{r,n}}{4\pi} + (1 - f_n) \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{Q_m}{4\pi} + \frac{Q_r}{4\pi}, \quad (24a)$$

$$\frac{\partial U_m}{\partial t} + f_n \sigma_{p,n} c U_{r,n} = \int_0^\infty \int_{4\pi} f_n \sigma_n I d\Omega' d\nu' + f_n Q_m, \quad (24b)$$

which are subject to the same initial and boundary conditions provided in Eqs. (1). In this form, Eq. (24a) can be solved using standard transport techniques to obtain an estimate of $\bar{I}$, after which the material energy at $t = t_{n+1}$ can be computed from the time-average of Eq. (24b),

$$U_{m,n+1} = U_{m,n} - \Delta t f_n \sigma_{p,n} c U_{r,n} + \int_0^\infty \int_{4\pi} f_n \sigma_n \bar{I} d\Omega' d\nu' + f_n \bar{Q}_m. \quad (25)$$

The temperature at t_{n+1} can then be determined from $U_{m,n+1}$ by inverting

$$U_{m,n+1} = U_{m,n} + \int_{T_n}^{T_{n+1}} c_v(T') dT'. \quad (26)$$

We emphasize that Eq. (23) is an auxiliary equation for calculating $\bar{I}$ from Eq. (24a) and T_{n+1} from Eq. (26); it is *not* equivalent to calculate $T_{n+1} = (U_{r,n+1}/a)^{1/4}$ unless $\beta(t)$ truly is constant.

The central feature of Eqs. (24) is the transformation of photon absorption and re-emission to an effective scattering treatment in which f_n is the absorption probability upon collision, and $(1 - f_n)$ is the effective scattering probability. By the definition of f_n in Eq. (21), it is clear that increasing Δt also increases the effective scattering probability, implying that, for long time steps, absorption and re-emission events are increasingly simulated as effective scattering events. As $\Delta t \rightarrow 0$, then $f_n \rightarrow 1$, reducing the effective scattering ratio and permitting the IMC equations to limit to the TRT equations at $t = t_n$. In Section 4 we address the consequences of these approximations on the temporal error of the IMC equations.

2.3. General Characteristics of the IMC Equations

Despite the three approximations made above – and we shall put these in context with other methodologies below – solutions of Eqs. (24) have several desirable features: they preserve the equilibrium solution, their energy conservation equation is identical to the TRT equations, and their solutions are generally non-negative. First, they preserve the equilibrium solution for any temperature T_0 , as $I(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = B(\nu, T_0)$ and $T(\mathbf{x}, t) = T_0$ reduces to

$$\sigma B(\nu, T_0) = (1 - f_n)\sigma B(\nu, T_0) + f_n\sigma B(\nu, T_0) = \sigma B(\nu, T_0), \quad (27a)$$

$$f_n\sigma_p a c T_0^4 = f_n\sigma_p a c T_0^4. \quad (27b)$$

The IMC equations also satisfy the same energy balance as the TRT equations, Eq. (15), which can be verified by integrating Eq. (24a) over all frequencies and energies, time-averaging, and using the definitions of E and $\mathbf{F}$ in Eqs. (12),

$$\frac{\Delta E}{\Delta t} + \nabla \cdot \mathbf{F} + f_n \int_0^\infty \sigma_n \bar{I} d\nu = f_n \sigma_{p,n} c U_{r,n} + (1 - f_n) \overline{Q_m} + \overline{Q_r} \quad (28)$$

Adding Eq. (28) to Eq. (25), the absorption-emission terms cancel, and the TRT energy conservation equation in Eq. (15) is reproduced exactly.

Because Eq. (24a) is a linear transport equation with positive sources, its solutions are guaranteed to be nonnegative (Case & Zweifel, 1967; Larsen & Mercier, 1987). This implies that the term $\int_0^\infty \int_{4\pi} f_n \sigma_n \bar{I} d\Omega d\nu$ in Eq. (25) is nonnegative. Does this imply that $T_{n+1} > 0$? That is, in the absence of sources and absorbed radiation in Eq. (25), we question whether

$$U_{m,n} - c f_n \sigma_{p,n} U_{r,n} \Delta t_n > 0? \quad (29)$$

Using the definition of β from Eq. (10), we write

$$\beta_n = \frac{4aT_n^3}{c_{v,n}} = 4 \frac{U_{r,n}}{c_{v,n} T_n}, \quad (30)$$

which may be solved for $U_{r,n}$ to produce

$$U_{r,n} = \frac{1}{4} \beta_n c_{v,n} T_n. \quad (31)$$

We therefore rewrite the inequality as (using Eq. (21)),

$$U_{m,n} - \frac{1 - f_n}{4\alpha} c_{v,n} T_n \stackrel{?}{>} 0, \quad (32)$$

and, since f_n is bounded below by 0, the following condition is sufficient for positivity:

$$U_{m,n} \stackrel{?}{>} \frac{1}{4\alpha} c_{v,n} T_n. \quad (33)$$

Since $\alpha \geq 1/2$ (and usually $\alpha = 1$), the question of whether this equation is satisfied primarily depends upon the definition of the specific heat $c_v(T)$. It is possible to find functional forms of the specific heat for which this inequality fails to hold. For instance, let us assume that the vast majority of physically meaningful specific heats may be adequately described by a monomial function,

$$c_v(T) = vT^z, \quad (34)$$

where v is a constant and $z > 0$ (note that for a perfect gas, $z = 0$, and for the linear ‘‘Su-Olson’’ problem described in (Su & Olson, 1997), $z = 3$). Then Eq. (33) becomes

$$\begin{aligned} v \frac{T_n^{z+1}}{z+1} &\stackrel{?}{>} \frac{1}{4\alpha} v T_n^{z+1}, \\ 1 &\stackrel{?}{>} \frac{z+1}{4\alpha}. \end{aligned} \quad (35)$$

In practice, the choice $\alpha = 1$ is almost always made. In that case, the inequality in Eq. (35) is true when $z \leq 3$. Since $z = 0$ for an ideal gas, and it is unlikely that $z > 3$ describes any physically meaningful material, and it is also likely that some radiation energy will be tallied in Eq. (25), we conclude that in most practical situations, Eq. (25) produces positive temperatures.

2.4. Alternative Methodologies and Context for the IMC Equations

The three approximations made to derive the IMC equations from the TRT equations serve as points of departure between it and competing methodologies: the linearization, the time-averaging of U_r using the parameter α , and the substitution of time-averaged unknowns with their time-continuous counterparts. We next briefly discuss some alternative approaches.

2.4.1. Alternative linearizations

Although nearly all methodologies make the time-explicit approximations for the temperature-dependent data in Eqs. (16), not all treatments introduce β in order to linearize the equations in U_r (T^4). For general problems, the linearization (and not the ensuing temporal discretization) has the leading order effect on solution accuracy, particularly in the equilibrium diffusion limit (Densmore & Larsen, 2004; Knoll et al., 2007; Larsen et al., 2013).

For example, the Ahrens-Larsen equations introduce a factor $\eta \equiv U_m/U_r$ and freeze this quantity at $t = t_n$; this linearizes the system in U_m and produces the coupled system,

$$\frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I = b_n \sigma_n \frac{c \eta_n U_m}{4\pi} + \frac{Q_r}{4\pi}, \quad (36a)$$

$$\frac{\partial U_m}{\partial t} + \sigma_n \eta_n c U_m = \int_0^\infty \int_{4\pi} \sigma_n I d\Omega d\nu + Q_m. \quad (36b)$$

These equations can be modeled in an analogous manner to neutron transport with ‘‘delayed’’ neutrons, and they are *exact* for linear problems (whereas IMC is not) (Ahrens & Larsen, 2001). Unfortunately, the innocuous-appearing Eqs. (36) can become temporally unstable for a relevant class of nonlinear problems (Wollaber & Larsen, 2011).

One criticism of the IMC equations is that their emission spectra are frozen at the previous time step due to the presence of b_n in Eq. (24a) (Byrne, 1973). A popular alternative in the deterministic transport community is to Taylor-expand the Planck emission term from its known value at t_n to t_{n+1} in an attempt to incorporate temporal derivative information in the emission spectra. Although the resulting linearized equations differ in this regard, their integrated radiation-matter energy exchange terms are identical, which implies that the bulk stability and accuracy characteristics for the IMC equations are broadly applicable to the deterministic community as well. We demonstrate this by implicitly time-discretizing Eqs. (11a) and

(11b) over a time step,

$$\frac{1}{c\Delta t}(I_{n+1} - I_n) + \mathbf{\Omega} \cdot \nabla I_{n+1} + \sigma_n I_{n+1} = \sigma_n B_{n+1} + \frac{\overline{Q_r}}{4\pi}, \quad (37a)$$

$$c_{v,n} \frac{\Delta T}{\Delta t} = \int_0^\infty \int_{4\pi} \sigma_n (I_{n+1} - B_{n+1}) d\Omega' d\nu' + \overline{Q_m}, \quad (37b)$$

where we have fixed σ and c_v at the beginning of the time step and defined $\Delta T = T_{n+1} - T_n$. Next, we Taylor-expand B_{n+1} from $t = t_n$,

$$B_{n+1} \approx B_n + \Delta T \frac{\partial B_n}{\partial T}, \quad (38)$$

and introduce it into Eq. (37b) in order to solve for ΔT ,

$$\begin{aligned} c_{v,n} \frac{\Delta T}{\Delta t} &= \int_0^\infty \int_{4\pi} \sigma_n \left(I_{n+1} - B_n - \Delta T \frac{\partial B_n}{\partial T} \right) d\Omega' d\nu' + \overline{Q_m}, \\ \Delta T &= \frac{\int_0^\infty \int_{4\pi} \sigma_n (I_{n+1} - B_n) d\Omega' d\nu' + \overline{Q_m}}{c_{v,n}/\Delta t + \int_0^\infty \int_{4\pi} \sigma_n \frac{\partial B_n}{\partial T} d\nu d\Omega}. \end{aligned} \quad (39)$$

We next define the Planck-derivative weighted opacity,

$$\sigma_R \equiv \frac{\int_0^\infty \int_{4\pi} \sigma_n \frac{\partial B_n}{\partial T} d\nu d\Omega}{\int_0^\infty \int_{4\pi} \frac{\partial B_n}{\partial T} d\nu d\Omega}, \quad (40a)$$

and note that

$$\int_0^\infty \int_{4\pi} \frac{\partial B_n}{\partial T} d\nu d\Omega = \frac{\partial}{\partial T} \int_0^\infty \int_{4\pi} B_n d\nu d\Omega = 4acT^3. \quad (40b)$$

Introducing Eqs. (40) into Eq. (39) and using the definition of β from Eq. (10),

$$\Delta T = \frac{\Delta t}{c_{v,n}} \frac{\int_0^\infty \int_{4\pi} \sigma_n (I_{n+1} - B_n) d\Omega' d\nu' + \overline{Q_m}}{1 + \sigma_R \beta c \Delta t} = \tilde{f}_n \frac{\Delta t}{c_{v,n}} \left(\int_0^\infty \int_{4\pi} \sigma_n (I_{n+1} - B_n) d\Omega' d\nu' + \overline{Q_m} \right), \quad (41)$$

where we have made an IMC-like definition of an alternative Fleck factor, $\tilde{f}_n$,

$$\tilde{f}_n \equiv \frac{1}{1 + \beta_n \sigma_{R,n} c \Delta t}. \quad (42)$$

We may now eliminate B_{n+1} in Eq. (37a) via Eq. (38) and Eq. (41),

$$\frac{1}{c\Delta t}(I_{n+1} - I_n) + \mathbf{\Omega} \cdot \nabla I_{n+1} + \sigma_n I_{n+1} = \sigma_n B_n + \sigma_n \frac{\partial B_n}{\partial T} \tilde{f}_n \frac{\Delta t}{c_{v,n}} \left(\int_0^\infty \int_{4\pi} \sigma_n (I_{n+1} - B_n) d\Omega' d\nu' + \overline{Q_m} \right) + \frac{\overline{Q_r}}{4\pi}. \quad (43)$$

We define the normalized derivative-emission term,

$$b'_n(\nu) \equiv \frac{\frac{\partial B_n}{\partial T}}{\int_0^\infty \frac{\partial B_n}{\partial T} d\nu} = \frac{\frac{\partial B_n}{\partial T}}{4acT_n^3/4\pi}, \quad (44)$$

and rearrange terms in Eq. (43) and (41) using the definitions of b_n , β_n , b'_n , σ_R , and $\tilde{f}_n$ from Eqs. (8), (10), (44), (40a), and (42) to arrive at the Taylor-expanded, linearized equations that are typically solved via deterministic methods,

$$\begin{aligned} \frac{1}{c\Delta t}(I_{n+1} - I_n) + \mathbf{\Omega} \cdot \nabla I_{n+1} + \sigma_n I_{n+1} &= \frac{\sigma_n b'_n}{\sigma_{R,n}} \frac{1}{4\pi} \int_0^\infty \int_{4\pi} (1 - \tilde{f}_n) \sigma_n I_{n+1} d\Omega d\nu \\ &+ \frac{\sigma_{p,n} c U_{r,n}}{4\pi} \left(\frac{\sigma_n b_n}{\sigma_{p,n}} - (1 - \tilde{f}_n) \frac{\sigma_n b'_n}{\sigma_{R,n}} \right) + \frac{\sigma_n b'_n}{\sigma_{R,n}} (1 - \tilde{f}_n) \frac{\overline{Q_m}}{4\pi} + \frac{\overline{Q_r}}{4\pi}, \end{aligned} \quad (45a)$$

$$c_{v,n} \frac{\Delta T}{\Delta t} + \tilde{f}_n \sigma_{p,n} c U_{r,n} = \int_0^\infty \int_{4\pi} \tilde{f}_n \sigma_n I_{n+1} d\Omega' d\nu' + \tilde{f}_n \bar{Q}_m. \quad (45b)$$

These equations are distinct from the IMC equations in that they incorporate the temperature-derivative of the Planck function, b'_n , and that the “Fleck factor” contains a frequency-integrated opacity with this weighting, σ_R instead of σ_p . However, to our knowledge, it is not widely known that the two approaches are identical for frequency-independent (gray) problems, for which $\sigma = \sigma_p = \sigma_R$ is a constant, in which case the “Fleck factors” in Eqs. (21) and (42) are identical. We demonstrate this by noting that the fractional terms involving b_n and b'_n are normalized to unity,

$$\int_0^\infty \frac{\sigma_n b_n}{\sigma_{p,n}} d\nu = \int_0^\infty \frac{\sigma_n b'_n}{\sigma_{R,n}} d\nu = 1,$$

so that the frequency-integrated form of Eq. (45a) reduces to

$$\begin{aligned} \int_0^\infty \left(\frac{1}{c\Delta t} (I_{n+1} - I_n) + \mathbf{\Omega} \cdot \nabla I_{n+1} + \sigma_n I_{n+1} \right) d\nu &= \frac{1}{4\pi} \int_0^\infty \int_{4\pi} (1 - \tilde{f}_n) \sigma_n I_{n+1} d\Omega d\nu \\ &+ \tilde{f}_n \frac{\sigma_{p,n} c U_{r,n}}{4\pi} + (1 - \tilde{f}_n) \frac{\bar{Q}_m}{4\pi} + \int_0^\infty \frac{\bar{Q}_r}{4\pi} d\nu, \end{aligned} \quad (46)$$

which is identical to the frequency-integrated, time-implicit form of the IMC radiation equation, Eq. (24a). This can be straightforwardly extended to show that the energy balance, Eq. (28) is also identical so long as $\sigma_R = \sigma_p$, which is always true for gray problems. We therefore conclude that the major difference between the time-implicit IMC equations and these is their treatment of the frequency emission term, which implies that stability and accuracy analyses of the gray forms of these equations are broadly applicable to either system. As a historical note, the primary reason that Monte Carlo solvers are not used to solve Eq. (45a) is that, although the frequency-integrated source strengths are identical, it is possible to find ν such that the $U_{r,n}$ emission term is negative, and typical Monte Carlo algorithms for particle transport preclude negative particle weights.

A more compelling alternative under active development is to rigorously treat the nonlinearity in the T^4 source by leaving it as an unknown quantity (Brooks, 1989; N’kaoua, 1991),

$$\frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I = b_n \sigma_n \frac{c U_{r,n+1}}{4\pi} + \frac{Q_r}{4\pi}, \quad (47a)$$

$$\frac{\partial U_m}{\partial t} + \sigma_n c U_r = \int_0^\infty \int_{4\pi} \sigma_n I d\Omega d\nu + Q_m. \quad (47b)$$

Monte Carlo particles with symbolic (unknown) energy-weights are then used to sample the unknown $U_{r,n+1}$ source with known angle and frequency information, and a collision-probability-like $J \times J$ coupling matrix is tallied, where J is the number of spatial zones. The matrix entries express the probabilities of particles born in zone i depositing some fraction of their energy into zone j . At the end of the transport calculation, this matrix can then be inverted as part of a nonlinear solve to determine T_{n+1} . This approach does contain temporal truncation errors due to using T_{n+1}^4 as the emission term and time-explicit data for the opacities and emission spectrum, but, because it rigorously treats the nonlinearity, it preserves the thick diffusion limit (Densmore & Larsen, 2004). This approach truly merits the moniker of “Implicit” Monte Carlo, although, because the name was already in common use, the methodology is referred to as “Symbolic Implicit Monte Carlo” (SIMC).

Finally, there are several other nonlinear approaches that use either fixed point or Newton iterations to converge T_{n+1} , either by using a decoupled, lower order method (such as gray diffusion) and post-processing with a Monte Carlo or other transport calculation, by rerunning the time-dependent Monte Carlo calculation, or by creatively coupling the low-order and high-order methods together (Heymann & Siebenmorgen, 2012; Iliev et al., 2006; Larsen et al., 2013; Park et al., 2012; Petkova & Springel, 2009).

2.4.2. Alternatives to IMC approximations in $U_r(t)$

A compelling criticism of the approximate treatment of $U_r(t)$ in Eq. (23) is that it is invalid at $t = t_n$ for $\alpha > 0$,

$$U_r(t = t_n) \neq f_n U_{r,n} + \frac{1}{c\sigma_{p,n}} \int_0^\infty \int_{4\pi} (1 - f_n) \sigma_n I_n d\Omega' d\nu' + \frac{1}{c\sigma_{p,n}} (1 - f_n) Q_m. \quad (48)$$

This results from the introduction of α and the replacement of time-average quantities with time-continuous counterparts. This can be remedied in several ways. Returning to Eq. (17c), after multiplying by β_n ,

$$\frac{\partial U_r}{\partial t} + \beta_n \sigma_{p,n} c U_r = \int_0^\infty \int_{4\pi} \beta_n \sigma_n I d\Omega' d\nu' + \beta_n Q_m, \quad (49)$$

we recognize this as a first-order differential equation in $U_r(t)$. Through the use of an integrating factor, this can be “solved” for $U_r(t)$,

$$U_r(t) = U_{r,n} e^{-\beta_n \sigma_{p,n} c(t-t_n)} + \int_{t_n}^t \beta_n e^{-\beta_n \sigma_{p,n} c(t-t')} \left(\int_0^\infty \int_{4\pi} \sigma_n I(t') d\Omega' d\nu' dt' + Q_m(t') \right) dt'. \quad (50)$$

Eq. (50) serves as the starting point for several alternative approaches, the oldest of which is the Carter-Forest methodology (Carter & Forest, 1973). The Carter-Forest equations can be constructed by substituting Eq. (50) into Eqs. (11a) and (11b),

$$\begin{aligned} \frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I &= \frac{\sigma_n b_n}{4\pi} c U_{r,n} e^{-\beta_n \sigma_{p,n} c(t-t_n)} \\ &+ \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \int_{t_n}^t \beta_n \sigma_{p,n} c e^{-\beta_n \sigma_{p,n} c(t-t')} \left[\int_0^\infty \int_{4\pi} \sigma_n I d\Omega' d\nu' + Q_m(t') \right] dt' + \frac{Q_r}{4\pi}, \end{aligned} \quad (51a)$$

$$\begin{aligned} \frac{\partial U_m}{\partial t} + \sigma_{p,n} c U_{r,n} e^{-\beta_n \sigma_{p,n} c(t-t_n)} &+ \int_{t_n}^t \beta_n \sigma_{p,n} c e^{-\beta_n \sigma_{p,n} c(t-t')} \left(\int_0^\infty \int_{4\pi} \sigma_n I(t') d\Omega' d\nu' dt' + Q_m(t') \right) dt' \\ &= \int_0^\infty \int_{4\pi} \sigma I d\Omega' d\nu' + Q_m. \end{aligned} \quad (51b)$$

It can be shown that the Carter-Forest equations are temporally exact for linear problems, that they are unconditionally stable and monotone, that the time-dependent source and effective scattering terms can be implemented efficiently, and that there are cases for which they preserve the equilibrium diffusion limit (Cheatham, 2010; Densmore & Larsen, 2004; Martin & Brown, 2002, 2001c; Wollaber & Larsen, 2011). Nevertheless, this methodology has not been widely implemented.

Alternatively, by treating the $I(t')$ term “implicitly” (substituting $I(t)$ for $I(t')$) and using the time-average value of Q_m , Eq. (48) can be further simplified,

$$U_r(t) = U_{r,n} e^{-\beta_n \sigma_{p,n} c(t-t_n)} + \frac{1}{\sigma_{p,n} c} \left(1 - e^{-\beta_n \sigma_{p,n} c(t-t_n)} \right) \left[\int_0^\infty \int_{4\pi} \sigma_n I(t) d\Omega' d\nu' + \overline{Q}_m \right]. \quad (52)$$

Defining $\hat{f}(t)$ as a time-dependent “Fleck factor”

$$\hat{f}_n(t) \equiv e^{-\beta_n \sigma_{p,n} c(t-t_n)}, \quad (53)$$

and substituting Eq. (53) and Eq. (52) into Eqs. (11a) and (11b), we obtain IMC equations with a time-dependent Fleck factor, the IMC-TDF equations (Wollaber & Larsen, 2009b),

$$\frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I = \frac{\sigma_n b_n}{4\pi} c U_{r,n} \hat{f}_n(t) + \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} (1 - \hat{f}_n(t)) \left[\int_0^\infty \int_{4\pi} \sigma_n I d\Omega' d\nu' + \overline{Q}_m \right] + \frac{Q_r}{4\pi}, \quad (54a)$$

$$\frac{\partial U_m}{\partial t} + \hat{f}_n(t) \sigma_{p,n} c U_{r,n} = \int_0^\infty \int_{4\pi} \hat{f}_n(t) \sigma_n I d\Omega' d\nu' + \hat{f}_n(t) Q_m. \quad (54b)$$

Eqs. (54) are identical to the IMC equations, Eqs. (24), except that $\hat{f}_n(t)$ has replaced f_n and obviated the need for the user-parameter α . The convenience of retaining a more IMC-like system of equations comes at a nominal cost in accuracy; Eqs. (54a) are $O(\Delta t^2)$ accurate as $\Delta t \rightarrow 0$ for linear problems (whereas Eqs. (51) are exact), although it is still necessary to modify the probabilities of scattering and emission times to account for $\hat{f}_n(t)$. Martin and Brown also explored this system, denoting them the “Analytic Monte Carlo-Implicit” equations (Martin & Brown, 2001b,c). They also considered an “AMC-Explicit” approach in which $I(t')$ is replaced with $I(t_n)$, making it less stable than the IMC approximations. Additionally, we note that if one replaces the exponential $\hat{f}_n(t)$ with its first-order Padé approximant (i.e., $e^{-z} \approx 1/(1+z)$), the resulting system is only conditionally stable (Wollaber, 2008).

One can make an additional approximation, as in the IMC equations, by approximating the time-average of U_r in Eq. (52),

$$\overline{U_r} \approx \overline{\hat{f}_n(t)} U_{r,n} + \frac{1}{c} (1 - \overline{\hat{f}_n(t)}) \bar{I}, \quad (55)$$

where the time-averaged value of $\hat{f}_n(t)$ is

$$\overline{\hat{f}_n(t)} \equiv \frac{1}{\beta_n \sigma_{p,n} c \Delta t} (1 - e^{-\beta_n \sigma_{p,n} c \Delta t}). \quad (56)$$

Setting this equal to the definition of the traditional Fleck factor in Eq. (21) and solving for α , we then determine a locally adaptive value of alpha,

$$\alpha(\Delta t) = \frac{1}{1 - e^{-\beta_n \sigma_{p,n} c \Delta t}} - \frac{1}{\beta_n \sigma_{p,n} c \Delta t}. \quad (57)$$

It can be shown that $\alpha(\Delta t) \rightarrow 0.5$ as $\Delta t \rightarrow 0$ and that $\alpha(\Delta t) \rightarrow 1$ as $\Delta t \rightarrow \infty$, which implies that the $O(\Delta t^2)$ “Cranck-Nicholson” behavior is automatically selected for short time steps and that the approximation becomes more implicit for large time steps (Wollaber, 2008, 2011; Wollaber & Larsen, 2009b). This also implies that modest accuracy gains can be had for problems employing sufficiently small time steps in existing IMC implementations by replacing the user-defined α with a locally defined $\alpha(\Delta t)$ in Eq. (57).

Finally, instead of replacing the time-average $\overline{U_r}$ with its instantaneous value, it has been suggested to use the end-of-time-step value $U_{r,n+1}$ in order to reduce the magnitude of the coupling in the absorption-emission physics (McClarren & Urbatsch, 2009),

$$U_{r,n+1} \approx e^{-\beta_n \sigma_{p,n} c \Delta t} U_{r,n} + \frac{1}{c} (1 - e^{-\beta_n \sigma_{p,n} c \Delta t}) I(t). \quad (58)$$

This suggests a Fleck factor given by

$$f_\infty \equiv e^{-\beta_n \sigma_{p,n} c \Delta t}, \quad (59)$$

however, because this decays extremely rapidly and can decouple the radiation from the material entirely, the following modification was suggested,

$$f_\ell \equiv f_\infty + \frac{1}{\ell} \frac{(\beta_n \sigma_{p,n} c \Delta t)^\ell}{(1 + \beta_n \sigma_{p,n} c \Delta t)^{\ell+1}}, \quad \ell > 2, \quad (60)$$

to allow f_ℓ to retain a slower, rationally decaying component. This user-control of the coupling can help prevent overheating for large time steps. Additionally, the adaptive α in Eq. (57) can be combined with Eq. (60) to produce a synthesized Fleck factor designed to retain the enhanced accuracy for asymptotically short and long time steps (Wollaber, 2011).

Although this is not an exhaustive list, it is hopefully clear that the IMC equations are a particular linearization and time-discretization of the TRT equations in a family of linearizations that represent absorption-reemission physics through effective scattering. This subsection should also demonstrate that several of the approximations made to arrive at the IMC equations can be eliminated or ameliorated at the expense of introducing more complicated, time-dependent terms into the final system.

2.5. An Overview of a Monte Carlo Implementation

In the previous section, the IMC equations were derived and put into context, but there was little discussion regarding their numerical solution. In this section we provide an overview of their Monte Carlo solution.

A *Monte Carlo* method uses stochastic techniques to estimate the solution of a problem. In general, such problems are simulated on a computer using a large number of random or pseudorandom numbers combined with probability density functions (PDFs) to generate an average result. Monte Carlo transport methods are also more closely linked to the transport processes of physical particles, which implies that the underlying functions and terms in their computational implementation usually retain clear, physical interpretations. By contrast, deterministic transport methods often require careful calculation and attribution of a multitude of terms that make up the coefficients of the underlying large, algebraic system, making them more tedious to implement correctly. Monte Carlo methods also require fewer approximations than deterministic methods; in linear Monte Carlo transport methods, for instance, there is no need to discretize the problem, provided that continuous problem parameters are known.

Generating a Monte Carlo method for a transport problem involves careful construction of PDFs based on the physical events a particle could undergo in its lifetime (sometimes referred to as a particle *history*). A generic particle history employs random numbers to create a particle inside the simulated problem geometry, choose where the particle travels, choose a collision or leakage event, etc., until the particle loses relevance (by leaking outside the problem, for instance). Typically, many such particles are simulated, and events of interest are tallied and averaged. At the method's conclusion, the average result is reported with some confidence estimate, usually the standard deviation of the average. It can easily be shown using the Central Limit Theorem that the standard deviation of the average decreases at a rate that is inversely proportional to the square root of the number of histories. That is, to improve the confidence in an estimate by a factor of two, one must on average simulate four times as many particles. Correspondingly, Monte Carlo methods have the disadvantage of generally being quite computationally expensive – it may take an astronomical number of histories to obtain a particular solution estimate with appreciable confidence. It is worth noting that Monte Carlo methods are so-called “embarrassingly parallel” algorithms with regard to the distribution of particles among independent processors, provided that the problem description fits into a single processor's memory. For such problems, it is almost trivial to convert a single-processor algorithm to run on very large, distributed computing systems. Monte Carlo methods are widely used, often giving researchers the powerful ability to use two wholly different methods (stochastic and deterministic) to estimate a problem solution.

Even if continuous problem data is available, it is not possible to exactly solve Eqs. (24) using a Monte Carlo method. This is fundamentally due to the nonlinearity in the material temperature, i.e., the effect that the material temperature has on the underlying problem data. The simplest remedy for this problem is to impose a spatial grid in which the temperature is defined to be a constant or some other spatially fitted, functional expansion. No matter how this is done, the imposition of some sort of spatial discretization introduces another source of error into the problem. Additionally, the numerical treatment of the absorption-reemission approximation can introduce an unphysical wavespeed in the problem. Although it is possible to store the locations of each absorbed Monte Carlo photon and construct a spatial temperature distribution within each zone, it is customary to assume either a histogram or linear temperature distribution. Thus, absorbed photons generally lose their identities in a spatial zone and are reemitted uniformly or linearly in space during the subsequent time step. If the spatial zone is large, then it is possible for a photon to be absorbed on one side of the zone and be released at the other side faster than an underlying radiation wavefront. In subsequent analyses, we shall assume that a spatial grid has been imposed on which the temperature and material data have some discrete representation with the understanding that this will introduce an additional source of error.

In general, one has to represent many physical photons with relatively few Monte Carlo photons due to computational resource limitations; Monte Carlo photons therefore represent large packets of physical photons. In TRT calculations, there is a slight, additional subtlety that arises – the Monte Carlo particle density is not proportional to the number density of the photons. However, this subtlety may be exploited to develop a compelling solution to the problem of local (per-Monte Carlo particle) energy conservation. Central to these ideas is the concept of a Monte Carlo particle *weight*.

To the reader more familiar with linear, neutron-transport Monte Carlo problems, a particle weight is a dimensionless quantity that represents the relative number of neutrons represented by a Monte Carlo particle. The particle weight is therefore a new independent variable that is controlled by the user and by the Monte Carlo tracking algorithm. In an *analog* neutron transport calculation, in which the particle weight is fixed, the Monte Carlo number density is directly proportional to the neutron number density. This is not the case in a frequency-dependent TRT calculation. The contrast between these two types of problems merits discussion, since we will extend the experience from the linear, neutron-transport Monte Carlo community to create a Monte Carlo interpretation of the IMC equations.

The central reason for the different treatments of the Monte Carlo particle weights is that the unknown for the radiation in a TRT problem, $I(\nu)$, is proportional to the frequency ν times the photon number density $N(\nu)$ (with the arguments $\mathbf{x}, \boldsymbol{\Omega}, t$ suppressed for clarity):

$$I(\nu) = h\nu c N(\nu).$$

The most straightforward Monte Carlo interpretation of Eq. (24a) tallies a quantity that is proportional¹ to $I(\nu)$, not $N(\nu) = I(\nu)/h\nu$. By comparison, let us artificially group the terms according to “neutronics” terminology,

$$\begin{aligned} I(\nu) &= (h\nu)[cN(\nu)], \\ I(\nu) &= \text{energy} \times \text{“angular flux”}. \end{aligned}$$

In a neutronics problem, the “angular flux”, which is the product of the neutron speed and number density, is the central unknown. Thus, any neutronics-like Monte Carlo interpretation of Eqs. (24) would introduce an “energy” term that depends upon the photon frequency. For reasons that will become clear later, this is undesirable. To aid this discussion, let us define, in a manner similar to (Ueki & Larsen, 1998),

$$M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) \, dV \, d\boldsymbol{\Omega} \, d\nu \, dw = \begin{array}{l} \text{The expected number density of Monte} \\ \text{Carlo particles at location } (\mathbf{x}, \boldsymbol{\Omega}, \nu) \\ \text{in } (dV, d\boldsymbol{\Omega}, d\nu) \text{ with weight } w \text{ in } dw \\ \text{traveling at time } t. \end{array} \quad (61)$$

In general, we would like to construct the intensity I from the Monte Carlo density M using the relationship

$$I(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = ch\nu N(\mathbf{x}, \nu, t) \equiv \int_0^\infty cw M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) \, dw, \quad (62)$$

which is to say,

$$h\nu N(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = \int_0^\infty w M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) \, dw. \quad (63)$$

The left side of this expression has units of energy per volume-angle-frequency. Since M is a number density, w cannot be dimensionless – w must have units of energy. As an example of how this weight could be treated, let us stipulate that $w = w_0$, a constant. Then, we may write M explicitly as

$$M_{w_0}(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) = M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) \delta(w - w_0), \quad (64)$$

¹For instance, in equilibrium, we should find the solution $I(\nu) = B(\nu)$. Monte Carlo algorithms for TRT treat this problem by sampling from $B(\nu)$ directly, not by sampling $B(\nu)/h\nu$. Therefore, $M(\nu)$ and $N(\nu)$ have different spectral distributions.

in which case we have

$$\begin{aligned}
h\nu N(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) &= \int_0^\infty w M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) \delta(w - w_0) dw, \\
&= w_0 M(x, \mu, \nu, t), \\
\int_0^\infty \int_V \int_{4\pi} h\nu N(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) d\Omega dV d\nu &= w_0 \int_0^\infty \int_V \int_{4\pi} M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) d\Omega dV d\nu, \\
\text{total system energy at time } t &= w_0 \times \text{total number of Monte Carlo particles}, \\
w_0 &= \frac{\text{energy}}{\text{Monte Carlo particle}}.
\end{aligned}$$

In other words, w_0 is the amount of energy that is represented by a single Monte Carlo particle. This type of Monte Carlo particle weight is sometimes distinguished from its linear, neutron-transport counterpart by terming it an *energy-weight*.

It is worthwhile to provide IMC equations in which w – the amount of energy represented by a Monte Carlo particle – is an independent variable. If we define

$$\tilde{I}(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) \equiv cw M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w), \quad (65)$$

so that

$$I(\mathbf{x}, \boldsymbol{\Omega}, \nu, t) = \int_0^\infty \tilde{I} dw, \quad (66)$$

the following Eqs. (67) are consistent with Eqs. (24),

$$\begin{aligned}
\frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \tilde{I} + \sigma_n \tilde{I} &= \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \iiint p(w' \rightarrow w) (1 - f_n) \sigma_n \tilde{I} d\Omega' d\nu' dw' \\
&+ f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{c U_{r,n}}{4\pi} \delta(w - w_0) + (1 - f_n) \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{Q_m}{4\pi} \delta(w - w_0) + \frac{Q_r}{4\pi} \delta(w - w_0), \quad (67a)
\end{aligned}$$

$$U_{m,n+1} = U_{m,n} - cf_n \sigma_{p,n} U_{r,n} \Delta t_n + \int_{t_n}^{t_{n+1}} \iiint f_n \sigma_{a,n} \tilde{I} d\mu d\nu dw dt. \quad (67b)$$

Here the triple integral denotes an integral over all directions, frequencies, and nonnegative weights. The quantity w_0 can be provided by the user or calculated by equally partitioning the radiation energy among a targeted number of Monte Carlo particles, or it can more generally be set to a zone-dependent quantity to allow for fewer or greater numbers of particles to be emitted in a particular zone with higher or lower energy-weights. The quantity $p(w' \rightarrow w)$ is the probability that a Monte Carlo particle with weight w' emerges with weight w after an effective scattering event. In the simplest scenario, in which constant particle weights are employed², this probability is given by:

$$p_{\text{constant}}(w' \rightarrow w) = \delta(w' - w), \quad (67c)$$

but other interpretations of p and Eqs. (67) may be formulated that correspond with certain variance reduction techniques; some of these are discussed below. Boundary and initial conditions to these equations are given by:

$$\tilde{I}(\mathbf{x}, \boldsymbol{\Omega}, \nu, 0, w) = I^i(\mathbf{x}, \boldsymbol{\Omega}, \nu) \delta(w - w_0), \quad (67d)$$

$$T(\mathbf{x}, 0) = T^i(\mathbf{x}), \quad (67e)$$

$$\tilde{I}(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) = I^b(\boldsymbol{\Omega}, \nu, t) \delta(w - w_0), \quad \mathbf{x} \in \partial V, \quad \boldsymbol{\Omega} \cdot \mathbf{n} < 0. \quad (67f)$$

²If the term “analog” is defined to mean “a numerical process that is completely analogous to the underlying physics”, then using constant particle weights is not strictly an analog procedure, since Monte Carlo photons used to solve Eqs. (67) have energies that do not depend on their frequency. Regardless, “analog tracking” is often used interchangeably with constant particle weight tracking.

These equations, when operated upon by $\int_0^\infty (\cdot) dw$, reduce to the traditional form of the IMC equations given by Eqs. (24). However, Eqs. (24) by themselves do not have a straightforward Monte Carlo interpretation, whereas Eqs. (67) do. It should be noted that Eqs. (67) are not unique; one may define alternative Monte Carlo interpretations of Eqs. (24) (for instance, one might attempt to define a Monte Carlo scheme that is completely analogous to the physics of photon transport). Yet, Eqs. (67) are typical of what one encounters in existing implementations. An additional advantage of Eqs. (67) is that energy conservation on a per-particle basis becomes trivial. From these equations, we see that if Eq. (67c) is used, the energy-weight of a particle never changes, *even if the particle frequency changes*. That is, a particle’s energy-weight is *not* related to its frequency, even though $h\nu$ is the traditional photon energy. In Eqs. (67), ν is simply treated as a parameter, and a Monte Carlo procedure used to solve these equations is not analogous to the physics of photon transport.

Thus, particle energy is introduced into the problem and travels through the system until it is either absorbed in the material, leaks out of the system, or goes to census; it can never be lost. Certain transport schemes in which the energy-weight is allowed to fluctuate may also be developed to contain this property, and will be discussed later. Finally, we note that if we had developed a Monte Carlo interpretation of the gray version of the IMC equations, it would not have been necessary to introduce the concept of energy-weights. However, had we done so, we would have had to perform a separate analysis for the frequency-dependent problem and introduce the concept there. Also, if one later desired to create a non-constant energy-weight kernel $p(w' \rightarrow w)$ to reduce the variance of the Monte Carlo process, it would again be necessary to introduce some type of weight to the gray equations.

3. Implicit Monte Carlo Implementation

In this section we describe details that are necessary to computationally implement a constant energy-weight version of Eqs. (67), the simplest numerical scenario possible. This section is intentionally basic, and the novice reader may benefit from reading this in conjunction with other, more general texts on Monte Carlo particle transport (Brown & Sutton, 1996; Haghighat, 2015; Kalos & Whitlock, 1986; Lewis & Miller Jr., 1984). Readers already proficient in Monte Carlo transport may primarily identify how existing techniques for linear transport are applied to IMC and need not consider this section in great detail.

To solve Eqs. (67) using a Monte Carlo method, it helps to pause and consider the possible energy sources and sinks of physical photons. For instance, in the first time step, Monte Carlo photons may be “born” from the initial condition (existing radiation), the boundary condition, the inhomogeneous source Q , or the material if $T > 0$. At the end of the time step, some of these photons may continue to exist in the radiation – these are generally stored in a *census* to preserve their state in preparation for the next time step. Other photons may have been absorbed in the material during the time step – their energy is subsequently removed from the radiation and attributed to the temperature. At the beginning of the next time step, photons may then emerge from the census (which is like an initial condition), the boundary condition, Q , or be emitted from the material. Using this as a guide, a natural algorithm emerges to begin the sampling procedures. This process is delineated explicitly below.

3.1. Simple Particle Sourcing

First, we discuss the various particle-energy sources in an IMC calculation. The total energy emitted by an isotropic, inhomogeneous source Q is

$$E_Q = \int_{t_n}^{t_{n+1}} \int_V \int_0^\infty Q(\mathbf{x}, \nu, t) d\nu dV dt. \quad (68)$$

Here, the particle time and frequency are chosen according to the functional form of Q . The particle angle is uniformly chosen in the unit sphere – in practice this is typically done by choosing μ (the cosine of the polar angle) uniformly on $[-1, 1]$, or

$$\mu = 2\xi_1 - 1, \quad (69a)$$

where ξ_i ($i \in \mathbb{N}$) is a *uniform random variate* (URV) on $(0, 1]$, and

$$\phi = \xi_2 2\pi, \quad (69b)$$

and constructing $\mathbf{\Omega}$ using

$$\mathbf{\Omega} = \left(\mu, \sqrt{1 - \mu^2} \cos \phi, \sqrt{1 - \mu^2} \sin \phi \right). \quad (70)$$

or similar, depending on the desired coordinate representation of solid angle. A uniform random variate is a particular number chosen from a uniform distribution of pseudorandom numbers. Modern computers typically allow one to obtain large streams of ξ_i with great efficiency. In what follows, choosing $\mathbf{\Omega}$ using Eqs. (69) is what is meant by “sampling isotropically in angle.”

To choose a particle location, one first observes that typical geometry models of transport problems consist of many non-overlapping, connected spatial zones. In general, the problem data is decreed to be constant or linear over each spatial zone. Here, we consider constant data over orthogonal spatial zones, but it is not difficult to extend this to arbitrarily oriented spatial zones with linear shape functions by constructing the proper PDF and sampling it through rejection or inversion from the cumulative distribution function (CDF) (Devroye, 1986).

Given w_0 in each zone (or a targeted number of particles for each zone), a Monte Carlo particle sourcing algorithm may begin to source particles by marching through each zone and determining the number of particles to create. As an example of choosing a spatial location within a zone uniformly, consider an orthogonal zone given by $\mathbf{x} \in V_j$, where V_j is a box described by $x \in [x_{j,\text{left}}, x_{j,\text{right}}]$, $y \in [y_{j,\text{left}}, y_{j,\text{right}}]$, and $z \in [z_{j,\text{left}}, z_{j,\text{right}}]$. In this case, a location is selected according to

$$\mathbf{x} = (x_{j,\text{left}} + \xi_3(x_{j,\text{right}} - x_{j,\text{left}}), y_{j,\text{left}} + \xi_4(y_{j,\text{right}} - y_{j,\text{left}}), z_{j,\text{left}} + \xi_5(z_{j,\text{right}} - z_{j,\text{left}})). \quad (71)$$

The particle’s initial energy-weight is set to the user-specified value of w_0 for this energy source for all the remaining particles in zone j , and the sequence is repeated for all cells containing a nonzero source Q . To set the particle’s initial frequency, one constructs a PDF by normalizing the frequency-dependence of Q . This can be done by determining the average zonal frequency dependence $Q_j(\nu)$ from

$$Q_j(\nu) \equiv \frac{1}{V_j \Delta t} \int_{V_j} \int_{t_n}^{t_{n+1}} Q(\mathbf{x}, \nu, t) dV dt, \quad (72)$$

and then determining the zonal frequency-dependent PDF from

$$f_{Q_j}(\nu) d\nu \equiv \frac{Q_j(\nu)}{\int_0^\infty Q_j(\nu) d\nu} d\nu. \quad (73)$$

It is also possible to consider more complicated sub-zone treatments of sampling the frequency-dependence according to a spatial or temporal shape (that is, by first sampling a position and then sampling ν using the conditional probability that $\mathbf{x}$ was chosen), although this is more computationally demanding. The particle’s initial time may be sampled similarly by replacing ν with t in the above procedure for sampling frequency. For Q_j constant within a time step, the time is simply assigned using

$$t = t_n + \xi \Delta t. \quad (74)$$

The total energy due to the boundary source is obtained by integrating the boundary condition over the time step, boundary surface, and angle,

$$E_B = \int_{t_n}^{t_{n+1}} \int_{\partial V} \int_0^\infty \int_{\mathbf{\Omega} \cdot \mathbf{n} > 0} (\mathbf{\Omega} \cdot \mathbf{n}) I^b(\mathbf{x}, \mathbf{\Omega}, \nu, t) d\mathbf{\Omega} d\nu dS dt \quad (75a)$$

Hence, one chooses particle location (on the boundary), direction, frequency, and time according to the functional forms of I^b . However, the boundary source is frequently specified as a blackbody at a fixed temperature, in which case the total energy may be written more simply as

$$E_B = \frac{ac\Delta t}{4} \int_{\partial V} T^4(\mathbf{x}) dS, \quad (75b)$$

in which case the particle frequency is chosen from a Planckian spectrum³ at the boundary temperature. More frequently, boundary temperatures are specified less generally (for instance, a constant temperature on the external sides of boundary zones), in which case the energy is more simply

$$E_B = \frac{ac\Delta t}{4} \sum_j T_j^4 A_j, \quad (75c)$$

where A_j is the exposed side for zone j and T_j is the temperature of that side. Typically, the particle time is chosen uniformly over the time step as in Eq. (74), and the direction is isotropic in the incoming directions. This latter task is sampled by choosing Ω coordinates to align with $\mathbf{n}$ on a particular zonal boundary such that $\mu = \Omega \cdot \mathbf{n}$ is the angle-cosine of the entering radiation. Then the entering angle PDF can be constructed from

$$f_b(\mu)d\mu = \frac{\mu I^b(\mu) d\mu}{\int_0^1 \mu I^b(\mu) d\mu}, \quad (76)$$

which, for isotropic $I^b(\mu)$ is

$$f_b(\mu)d\mu = 2\mu d\mu. \quad (77)$$

The procedure to invert the CDF and obtain μ is therefore

$$\begin{aligned} \int_0^\xi d\xi' &= \int_0^\mu 2\mu' d\mu', \\ \xi &= \mu^2, \\ \mu &= \sqrt{\xi}. \end{aligned} \quad (78)$$

The particle angle can then be constructed by sampling ϕ from Eq. (69b), writing Ω from Eq. (70) and rotating the Ω coordinate system back to the original problem if necessary. Particle energy-weights w should be assigned such that their sum accounts for all energy entering a local zone j and such that all energy-weights sum to E_B .

During the first time step, energy may be included due to the initial condition on I (in subsequent time steps, one simply “follows” the existing census particles):

$$E_{IC} = \frac{1}{c} \int_{4\pi} \int_V \int_0^\infty I^i(\mathbf{x}, \Omega, \nu) d\nu dV d\mu. \quad (79)$$

Here, particle location is chosen zone-wise and then within the zone as is done for Q , while direction and frequency depend on the form of the initial condition. The initial condition is also frequently given as a blackbody at a certain temperature, in which case the particle direction is isotropically sampled and particle frequency is chosen from a Planckian spectrum. The only notable difference from sampling for Q is that the particle time is specified at $t = 0$.

Finally, we turn to particles emitted from the material during the time step, which is represented by the first term on the right side of Eq. (67a). This time-independent term represents radiative energy held by the material at the beginning of the time step and radiated uniformly in t during the time step. We rewrite it here to analyze its features,

$$f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{c U_{r,n}}{4\pi} \delta(w - w_0). \quad (80)$$

The factor $1/4\pi$ tells us that this is an isotropic photon source. The factor $\sigma_n(\nu)b_n(\nu)/\sigma_{p,n}$ may be thought of as the frequency distribution term, similar to a fission spectrum $\chi(E)$ in neutron transport. The factor $\delta(w - w_0)$ specifies the initial particle energy-weight. The remaining quantities are what is left after integrating over

³ Fleck & Cummings provide details of how to computationally sample a Planckian using only uniform random variates (Fleck & Cummings, 1971).

frequency and angle – the source strength in units of energy per volume-time. The total particle energy injected during the time step due to this term is therefore

$$\begin{aligned} E_R &= \int_{t_n}^{t_{n+1}} \int_V \int_0^\infty \int_{4\pi} \frac{1}{4\pi} \frac{\sigma_n b_n}{\sigma_{p,n}} c f_n \sigma_{p,n}(\mathbf{x}) U_{r,n}(\mathbf{x}) d\Omega d\nu dV dt, \\ &= c \Delta t_n \int_V f_n \sigma_{p,n}(\mathbf{x}) U_{r,n}(\mathbf{x}) dV. \end{aligned} \quad (81)$$

The particle's location can be determined on a zone-wise basis in a similar manner as was done for Q , and it is emitted uniformly during the time step according to Eq. (74) and Eq. (69) since E_R has no temporal or angular dependence. We note that using and sampling from a linear (or better) spatial profile of $U_{r,n}$ is typically necessary for spatial accuracy; see Section 5.

The total energy introduced into the problem during a time step is therefore given by

$$E_{TOT} = E_Q + E_B + E_{IC} + E_R, \quad (82)$$

where E_{IC} may be due to a census or the initial condition. After each particle is created, it is then necessary to track it through the relevant problem domain, which is discussed next.

3.2. Simple Particle Tracking

After a Monte Carlo particle energy source is selected and the particle is introduced into the problem geometry, the next task is to track it through the problem, updating the independent variables along the way. There are essentially three events that must be considered that affect the nature of a particle's flight through the system: (i) the particle could collide with an electron in an atom, (ii) the particle could exit the current region (and enter a region with different opacities), or (iii) the particle time could reach the end of time step value t_{n+1} . Associated with these three possibilities are three distances – a distance to collision, distance to the spatial boundary, and the distance that would be traveled until $t = t_{n+1}$. The calculation of these second two quantities is non-stochastic and straightforward. The distance to the next spatial boundary d_G , is the smallest positive distance in zone j for a particle at location $\mathbf{x}_0$ that satisfies

$$\mathbf{x}_{b,j} = \mathbf{x}_0 + \boldsymbol{\Omega} d_G, \quad \mathbf{x}_{b,j} \in \partial V_j, \quad (83a)$$

where $\mathbf{x}_{b,j}$ is the boundary location in direction $\boldsymbol{\Omega}$. The distance to the next time boundary, d_T is,

$$d_T = c(t_{n+1} - t). \quad (83b)$$

The distance to the next collision, d_C , is determined in the following way. Assume that at location s we have a density of photons given by $N(s)$. We then consider the incremental removal due to collisions along a flight path at a location $s + \delta s$. Using the definition of total opacity, this is

$$N(s + \delta s) = N(s) - \sigma \delta s N(s) + O(\delta s^2).$$

That is, to first order, the amount removed in a differential distance δs is the probability of interaction per distance, times the distance, times the original density. Rearranging and allowing $\delta s \rightarrow 0$,

$$\frac{dN}{ds}(s) = -\sigma N(s).$$

This ordinary differential equation has the solution

$$N(s) = C e^{-\sigma s},$$

where C is an unspecified constant. We next normalize this solution on $[0, \infty)$ (the entire domain of a particle's potential flight without collisions) to construct the PDF for the probability that a particle will travel a distance s before experiencing a collision in ds about s ,

$$N(s) = \sigma e^{-\sigma s}. \quad (83c)$$

This is the PDF that we must sample to determine d_C . Being exponential, it has the important property that it is a *memoryless* distribution (Wikipedia contributors, 2007). This essentially means that the probability of removal during flight over a given interval does not change if that interval is moved. This has the following Monte Carlo implication. Let us imagine that d_C is sampled and determined to be some large distance, many spatial zones away. Then it is equivalent to move particles to the zone boundaries, halt, and sample for d_C repeatedly, rather than attempt to move particles through a multitude of boundaries to the collision location. We next observe that since all CDFs are monotonically increasing on $[0, 1]$, and since our URVs are uniformly distributed on $[0, 1]$, we may construct a relationship between the URV ξ and the CDF:

$$\begin{aligned}\int_0^\xi d\xi' &= \int_0^{d_C} \sigma e^{-\sigma s} ds, \\ \xi &= -e^{-\sigma d_C} + 1, \\ d_C &= \frac{-\ln(1 - \xi)}{\sigma}, \\ d_C &= \frac{-\ln(\xi)}{\sigma},\end{aligned}\tag{83d}$$

where the last equality results from the probabilistic equivalence of $1 - \xi$ and ξ . The next step in the Monte Carlo method, then, is to calculate d_C , d_G , and d_T according to Eqs. (83), determine the minimum quantity, and either move the particle to the collision location, spatial boundary, or time boundary respectively. We next consider what to do for each of these events.

If the minimum distance d is to either of the spatial or time boundaries, the particle direction, energy-weight⁴, and frequency remain unchanged. The particle location and time is simply updated to those of the boundary using:

$$\mathbf{x}' = \mathbf{x} + \boldsymbol{\Omega}d, \tag{84a}$$

$$t' = t + d/c. \tag{84b}$$

If d_G was used, then we check to see whether the particle has leaked out of the boundary. If it has not, the sampling process begins again in the new spatial zone with appropriately updated material properties. If d_T was selected, then the particle “goes to census,” where it is stored in computer memory for the calculation of the next time step.

If the minimum distance is to collision, the particle data are also updated according to Eqs. (84), but the collision event must then be decided. To see how this is done, let us break up the collision removal term into two component pieces:

$$\sigma_{a,n}I = [f_n + (1 - f_n)]\sigma_nI \tag{85}$$

We can interpret the bracketed term as a discrete probability distribution for two separate collision events: effective absorptions and effective scatters. Therefore, with probability f_n , the particle undergoes an effective absorption – the particle’s energy is deposited at the collision location and no longer followed (Note that there is no source term in Eq. (67a) due to effective absorptions at the instantaneous time; absorptions appear as a source in Eq. (67b)). With probability $1 - f_n$, the particle undergoes an effective scatter. The reemission process is represented by the in-scatter source term of Eq. (67a). We rewrite it as

$$\text{in-scatter source} = \frac{1}{4\pi} \frac{\sigma_n b_n}{\sigma_{p,n}} \iiint p(w' \rightarrow w)(1 - f_n)\sigma_n \tilde{I} d\Omega' d\nu' dw' \tag{86}$$

In this term, the triple integral over $\sigma_{a,n}I$ is the collision rate due to all absorptions at location x and time t . The factor $(1 - f_n)$ is the (frequency-independent) probability that such an event may be interpreted as an effective scatter. As above, the factor $1/4\pi$ is an isotropic, angular redistribution term, and the factor

⁴In (Fleck & Cummings, 1971), a continuous energy deposition model is used; we discuss this in Section 6.3.

$\sigma_n b_n / \sigma_{p,n}$ is the emerging photon frequency spectrum. A new particle weight may be chosen according to $p(w' \rightarrow w)$; in the constant energy-weight case, the weight remains unchanged. After either an effective absorption or scatter has been selected, the particle is said to have collided, and, if the particle was not absorbed, the process of choosing a new distance is repeated.

Particle tracking ends when the particle has leaked out of the system, been absorbed, or gone to census. The entire process (birth, tracking, death) is generally repeated a large number of times to generate meaningful statistics. These statistics are generated via a “tallying” mechanism. Upon different particle events (collision, boundary-crossing, etc.), one updates a computer memory structure, or *tally* using information about the current particle. This information can be used to ensure energy conservation for verification purposes, to update the material properties to end-of-time-step values in preparation for the next time step, and to provide users with the problem solutions.

3.3. Tallies

The first several tallies we will examine are related to the radiation intensity. In most of the below tallies, we will consider events that occur in a single geometric zone with volume V_{zone} over a range of particle histories $i \in [1, 2, \dots N_{\text{hist}}]$. From the material update equation (67b), we immediately see that we need an estimate of the energy deposited in a zone due to the effective absorptions. We first note that

$$f_n \sigma_n(\mathbf{x}, \nu) c w M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) d\Omega d\nu dV dw = \begin{array}{l} \text{absorption energy deposition rate} \\ \text{in } d\Omega d\nu dV dw \text{ about } (\mathbf{x}, \boldsymbol{\Omega}, \nu, w) \\ \text{at time } t. \end{array}$$

We would like to accumulate this differential quantity over all frequencies, angles, and the time step; and average it over a zone volume to find

$$\begin{aligned} & \text{mean energy deposited due to all effective absorptions in } V_{\text{zone}} \\ &= \frac{1}{V_{\text{zone}}} \int_{V_{\text{zone}}} \int_{t_n}^{t_{n+1}} \iiint f_n \sigma_n(\mathbf{x}, \nu) c w M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) d\Omega d\nu dt dV dw, \\ &\approx \frac{1}{V_{\text{zone}}} \sum_{i=1}^{N_{\text{hist}}} w_i \equiv T_A \quad (87) \end{aligned}$$

where w_i is the total Monte Carlo particle energy deposited in V_{zone} due to *all* effective absorption collisions in history i during time step n . In other words, when a particle undergoes an effective absorption according to Eq. (85), we deposit its entire energy w_i into the current zone’s material⁵. We will use T_A as shorthand for the “effective absorption tally.”

We next turn to an estimate of the energy density during a particular point in time, or at a “snapshot.” This essentially follows immediately from the definition of the intensity:

$$\begin{aligned} & \text{volume-averaged energy density at time } t, E(t) \\ &= \frac{1}{V_{\text{zone}}} \int_{V_{\text{zone}}} \iiint w M(\mathbf{x}, \boldsymbol{\Omega}, \nu, t, w) d\Omega d\nu dV \approx \frac{1}{V_{\text{zone}}} \sum_{i=1}^{N_{\text{hist}}} w_i, \quad (88) \end{aligned}$$

where c is the speed of light and w_i is the energy of a Monte Carlo particle in the radiation at time t . Another quantity of interest is the partial flux, $F^\pm$, which provides the gross rate of energy flowing out of

⁵This procedure assumes that a constant energy-weight is being used. It is also possible to employ a technique known as *absorption weighting* or *survival biasing*, in which a fraction of the particle energy is deposited and the remaining fraction is retained in the radiation.

(F^+) or into (F^-) a surface with area A and surface normal $\mathbf{n}$. The partial fluxes may be used to check energy conservation or to obtain leakages across the problem boundary. The partial flux tally is

rate of energy traveling in direction $\mathbf{n}$ through area $A = F^+$

$$\begin{aligned} &= \frac{1}{\Delta t_n} \int_{t_n}^{t_{n+1}} \int_{\Omega \cdot \mathbf{n} > 0} \iint \Omega \cdot \mathbf{n} w M(\mathbf{x}, \Omega, \nu, t, w) d\nu dw d\Omega dt \\ &\approx \frac{1}{A \Delta t_n} \sum_{i=1}^{N_{\text{hist}}} w_i^+, \quad \Omega \cdot \mathbf{n} > 0, \end{aligned} \quad (89)$$

where w_i^+ is the total Monte Carlo particle energy that crosses area A during the time step (in the positive direction). If desired, we may combine this with another tally for F^- , the average energy going the opposite direction, and calculate an estimate of the net flux (current) F in direction $\mathbf{n}$:

$$\mathbf{F} \cdot \mathbf{n} = F \approx \frac{1}{A \Delta t_n} \left(\sum_{i=1}^{N_{\text{hist}}} w_i^+ - \sum_{i=1}^{N_{\text{hist}}} w_i^- \right) \quad (90)$$

where w_i^- is the total Monte Carlo particle energy that crosses area A during the time step (in any negative direction, dictated by $\Omega \cdot \mathbf{n} < 0$). Note that all of the above quantities may also be broken up into different frequencies – that is, instead of accumulating all of the particles into a single tally regardless of frequency, we may choose to segregate the particles into a multigroup structure.

It is now possible to rewrite the material update equation [Eq. (67b)] in terms of Monte Carlo tallies and sources. Using Eq. (87) and a zonal version of (81) in Eq. (67b), it is

$$U_{m,n+1} = U_{m,n} - \frac{E_{r,\text{zone}}}{V_{\text{zone}}} + T_A. \quad (91)$$

Once $U_{m,n+1}$ is known, the temperature T_{n+1} may be calculated through the relation $\partial U_m / \partial T = c_v(T)$.

We next turn to the problem of local energy conservation. We begin by integrating Eq. (13) over a spatial zone, frequency, and the time step to find (exactly),

$$\begin{aligned} &[E_0(t_{n+1}) - E_0(t_n)] + U_m(t_{n+1}) - U_m(t_n) \\ &= \int_{t_n}^{t_{n+1}} \left(\int Q_m(\mathbf{x}) dV + \iint Q_r(\mathbf{x}, \nu, t) dV d\nu \right) dt - \int_{t_n}^{t_{n+1}} \iint \nabla \cdot \mathbf{F}(\mathbf{x}, \nu, t) dV d\nu dt. \end{aligned} \quad (92)$$

This equation may be broken up into several terms. The first term concerns the radiative energy density at the “snapshot” time $t = t_{n+1}$, for which we may use Eq. (88). The second term is assumed known from the initial condition (or census). The third and fourth terms we may calculate using Eq. (91). The first term on the right side is just $E_{Q,\text{zone}}$, the energy due to inhomogeneous sources in a single zone that was simulated as in the discussion around Eq. (68). Finally, the last term may be rewritten using the current tallies defined by Eqs. (89) and (90),

$$\begin{aligned} \int_{t_n}^{t_{n+1}} \int_0^\infty \int_{V_{\text{zone}}} \nabla \cdot \mathbf{F}(\mathbf{x}, \nu, t) dV d\nu dt &= \int_{t_n}^{t_{n+1}} \int_0^\infty \int_{\partial V_{\text{zone}}} \mathbf{n} \cdot \mathbf{F}(\mathbf{x}, \nu, t) dS d\nu dt, \\ &= \sum_{A_i} A_i \mathbf{F} \cdot \mathbf{n}_i, \end{aligned}$$

where the latter sum is over all faces of the zone under consideration. Putting these results together, we see that energy is conserved if

$$V_{\text{zone}} [E_0(t_{n+1}) - E_0(t_n)] = c f_n \sigma_{p,n} U_{r,n} \Delta t_n V_{\text{zone}} + E_{Q,\text{zone}} V_{\text{zone}} - T_A V_{\text{zone}} - \sum_{A_i} A_i \mathbf{F} \cdot \mathbf{n}_i. \quad (93)$$

That is, the difference in the zonal radiation energy at the beginning and end of the time step is due to the gain terms from the material radiation, inhomogeneous source, and incoming partial currents, and the loss terms resulting from effective absorptions and leakages. All of the incoming quantities are explicitly controlled by the algorithm, and all of the energy-loss quantities can be accounted for via tallying.

3.4. Gray IMC Equations with Weights

We conclude our discussion of IMC implementations by providing the gray version of the IMC equations. If one can assume that the opacities are frequency-independent, then it is possible to integrate Eqs. (67) over all frequencies to obtain a gray version of the IMC equations. In that case, the equations become:

$$\begin{aligned} \frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \tilde{I} + \sigma_n \tilde{I} = & \frac{1}{4\pi} f_n \sigma_n c U_{r,n} \delta(w - w_0) \\ & + \frac{1}{4\pi} \int_{4\pi} \int_0^\infty p(w' \rightarrow w) (1 - f) \sigma_n \tilde{I} dw' d\Omega' + (1 - f_n) \frac{Q_m}{4\pi} \delta(w - w_0) + \frac{Q_r}{4\pi} \delta(w - w_0), \end{aligned} \quad (94a)$$

with the material update equation

$$U_{m,n+1} = U_{m,n} - c f_n \sigma_n U_{r,n} \Delta t_n + \int_{t_n}^{t_{n+1}} \int_{4\pi} \int_0^\infty f_n \sigma_n I dw d\Omega dt, \quad (94b)$$

and initial and boundary conditions given by:

$$\tilde{I}(\mathbf{x}, \boldsymbol{\Omega}, 0) = \tilde{I}^i(\mathbf{x}, \boldsymbol{\Omega}) \delta(w - w_0), \quad (94c)$$

$$T(\mathbf{x}, 0) = T^i(\mathbf{x}), \quad (94d)$$

$$\tilde{I}(\mathbf{x}, \boldsymbol{\Omega}, t, w) = I^b(\mathbf{x}, \boldsymbol{\Omega}, t) \delta(w - w_0), \quad \mathbf{x} \in \partial V, \boldsymbol{\Omega} \cdot \mathbf{n} < 0. \quad (94e)$$

The same mechanisms can then be used to generate Monte Carlo particles, track them, and tally them, but frequency information becomes superfluous. As noted earlier, it is not strictly necessary that Eqs. (94) contain an energy-weight parameter, however, it is expedient to provide it in the likely event that the user would like to employ a non-constant energy-weight procedure in order to reduce the variance of the Monte Carlo calculation.

3.5. Summary of the IMC Procedure

In this section we have presented sample implementation details for a Monte Carlo procedure to estimate the solutions of the IMC equations in order to provide the reader with an unambiguous reference. We did this with the intention of allowing the reader to more rapidly understand the central issues of the IMC approach than is presently possible from the current literature. The overall process can be summarized briefly as follows. To model Eqs. (24) using a Monte Carlo procedure that conserves energy, it is expedient to employ the concept of an energy-weight. This energy-weight is different from the photon energy, and the photon frequency ν is treated as a parameter. The energy-weight can then be explicitly included in the IMC equations [see Eqs. (67)] in order to provide them with a straightforward Monte Carlo interpretation. To solve these equations, we first calculate the amount of energy that must be represented by the radiation. This energy may come from the material emissions, an inhomogeneous source Q , the boundary condition, or the previous time step in the form of a census or the initial condition. We then allocate this energy to particles of equal energy-weight w_0 . Each particle is introduced into the problem by sampling its direction, position, frequency, and time from pertinent probability density functions. The particles then either travel to a zone boundary, undergo a collision, or enter the next census. If a collision occurs, the particle's energy-weight is either absorbed into the material or it scatters into a new direction and frequency. Energy is automatically conserved since a particle's energy-weight does not change, regardless of its frequency; the particle is either absorbed, leaks out of the system, or goes to census. Each particle is followed until its history has terminated, and the results are tallied and averaged. The tallies and “new” census then provide the algorithm with the necessary ingredients to update the temperatures and begin the calculation for the next time step.

4. Temporal Accuracy and Stability

In this section we consider the temporal accuracy of the IMC equations with respect to their asymptotic convergence to the TRT equations as $\Delta t \rightarrow 0$ and their robustness as Δt becomes large for infinite medium problems (no spatial variations) and for a fixed spatial grid. We shall also consider variations on the IMC equations that have been introduced that target enhanced accuracy in both of these regimes. This section does not address complications due to statistical noise in Monte Carlo simulations; we consider only the numerical properties of the (statistically converged) solutions to the IMC equations.

4.1. Short Time Step Accuracy

We begin our analysis of the short-time asymptotic accuracy of the IMC equations. There are two sources of error in the IMC equations that arise from the linearization and from the time-discretization. We can analyze these separately by first considering a particular choice of specific heat and opacity temperature-dependencies that result in *linear* TRT equations, namely that

$$c_v(T) = c_v T^3, \quad (95a)$$

$$\sigma(T) = \sigma, \quad (95b)$$

which implies that $\beta = 4a/c_v$ (a constant) by Eq. (10), so that $U_r = \beta U_m$ without approximation (Su & Olson, 1997). Under this assumption and assuming a 0-D, gray system, the TRT equations reduce to

$$\frac{dE}{dt} + \sigma c E(t) = \sigma c U_r(t), \quad (96a)$$

$$\frac{1}{\beta} \frac{dU_r}{dt} + \sigma c U_r(t) = \sigma c E(t). \quad (96b)$$

These equations can be solved by noting that the coefficient matrix has eigenvalues of 0 and $\gamma = \sigma c(\beta + 1)$ and (therefore) seeking a constant solution and exponential decay at $e^{-\gamma t}$. The solutions are, at the end of a time step $t_n \leq t \leq t_{n+1}$,

$$E_{n+1} = \frac{\beta E_n + U_{r,n}}{\beta + 1} + \frac{E_n - U_{r,n}}{\beta + 1} e^{-\gamma \Delta t}, \quad (97a)$$

$$U_{r,n+1} = \frac{\beta E_n + U_{r,n}}{\beta + 1} - \beta \frac{E_n - U_{r,n}}{\beta + 1} e^{-\gamma \Delta t}. \quad (97b)$$

Interestingly, although these problem data appear to be non-physical, Densmore and Larsen demonstrated that the resulting linear equations provide a first-order correction to the nonlinear TRT equations near equilibrium (Larsen & Densmore, 2003). Eqs. (97) depict a numerical solution with energy densities and temperatures that monotonically (exponentially) propagate to an equilibrium condition. Eqs. (97) are also the solutions of the Carter-Forest equations, Eqs. (51), as they contain no time-discretization errors (only linearization errors) (Cheatham, 2010; Martin & Brown, 2001c; Wollaber & Larsen, 2011). By contrast, the IMC equations under the assumptions in Eqs. (95) are:

$$\frac{dE}{dt} + f \sigma c E(t) = f \sigma c U_{r,n}, \quad (98a)$$

$$\frac{1}{\beta} \frac{dU_r}{dt} + f \sigma c U_{r,n} = f \sigma c E(t). \quad (98b)$$

Their solution is found similarly as a constant plus an exponential decay at a different rate $e^{-f \sigma c t}$,

$$E_{n+1}^{IMC} = U_{r,n} + (E_n - U_{r,n}) e^{-f \sigma c \Delta t}, \quad (99a)$$

$$U_{r,n+1}^{IMC} = U_{r,n} - \beta (U_{r,n} - E_n) (1 - e^{-f \sigma c \Delta t}). \quad (99b)$$

The residual error can then be found by subtracting Eqs. (99) from Eqs. (97). After collecting like terms, the residual error in the energy density is

$$R^{IMC} = \frac{U_{r,n} - E_n}{\beta + 1} (\beta + e^{-(\beta+1)\sigma c \Delta t} - (\beta + 1)e^{-f\sigma c \Delta t}), \quad (100)$$

and, Taylor-expanding the exponentials and f as $\Delta t \rightarrow 0$,

$$\begin{aligned} \lim_{\Delta t \rightarrow 0} e^{-(\beta+1)\sigma c \Delta t} &= 1 + (\beta + 1)\sigma c \Delta t - \frac{1}{2} ((\beta + 1)\sigma c \Delta t)^2 + O(\Delta t^3), \\ \lim_{\Delta t \rightarrow 0} e^{-f\sigma c \Delta t} &= 1 - \sigma c \Delta t + (\sigma c \Delta t)^2 \left(\alpha \beta + \frac{1}{2} \right) + O(\Delta t^3), \\ R^{IMC} &= (U_{r,n} - E_n) \left(\frac{1 - 2\alpha}{2} \beta (\sigma c \Delta t)^2 \right) + O(\Delta t^3), \end{aligned} \quad (101)$$

implying that the local truncation error in the IMC equations due to the time-discretization is $O(\Delta t^2)$ as $\Delta t \rightarrow 0$ unless $\alpha = 1/2$, in which case it is $O(\Delta t^3)$ (Cheatham, 2010; Mosher, 2006). This implies that the global truncation error (the error at an arbitrary time in the simulation) is $O(\Delta t)$ unless $\alpha = 1/2$, in which case it is $O(\Delta t^2)$, as the errors accumulate in time. Cheatham also demonstrates this in the presence of inhomogeneous sources (Cheatham, 2010), and goes on to demonstrate that both the Carter-Forest and IMC equations contain $O(\Delta t)$ truncation error for *nonlinear* problems. This is expected since σ and β are typically dependent on temperature (and, therefore, time), so that using their time-explicit values produces the $O(\Delta t)$ error (Cheatham et al., 2010a). Because that error can dominate the time-discretization errors, in practice there appears to be only a marginal advantage to using the more complicated Carter-Forest equations. In general, IMC accuracy is a larger concern in the “intermediate” to large Δt limit, as, in that regime, the IMC equations permit non-physical solutions.

4.2. Long Time Step Accuracy

The IMC equations permit two nonphysical deficiencies in their solutions for sufficiently large Δt : damped, temporal oscillations, and maximum principle violations. We also consider the behavior of the IMC equations in the asymptotic equilibrium diffusion limit, defined later in this section.

4.2.1. A Linearized Stability Analysis

It has been shown that the IMC equations (and the Taylor-expanded deterministic equations in Eqs. (45)) admit damped, oscillatory solutions (Mosher & Densmore, 2005; Wollaber & Larsen, 2011). The simplest way to demonstrate this mathematically is to linearize the 0-D IMC equations about an equilibrium solution; previous work indicated that the worst oscillations are for the slowest Fourier modes (Wollaber & Larsen, 2011). What follows is a generalization of an earlier stability analyses of the IMC equations that assumed the absorption opacity $\sigma \propto T^{-3}$ to provide an arbitrary monomial temperature dependence, or $\sigma \propto T^{-p}$. To do this, we non-dimensionalize the thermal radiative transfer (TRT) equations by writing them in terms of an arbitrary equilibrium condition given by temperature T_0 . We then apply approximations to these results to generate their corresponding, non-dimensionalized Implicit Monte Carlo (IMC) equations. Next, we use the conservation of energy to exactly solve the IMC equations for the scaled temperature from cycle to cycle. We then use a simple expansion to write cycle-to-cycle temperatures as perturbations away from the equilibrium solution, and develop an *amplification factor* ρ that geometrically relates the magnitude of the resultant temperature perturbations to the initial one, and to look for unstable ($|\rho| > 1$) and oscillatory modes ($\rho < 0$) in the solution. We find that the amplification factor generated by using an absorption opacity $\sigma \propto T^{-p}$ is identical to that generated previously for $\sigma \propto T^{-3}$, and conclude that all of the previous results (Wollaber, 2008; Wollaber & Larsen, 2011) carry over to the general case.

We begin from the 0-D, gray, thermal radiative transfer (TRT) equations with no scattering,

$$\frac{1}{c} \frac{dI}{dt} + \sigma I = \frac{1}{4\pi} c \sigma U_r, \quad (102a)$$

$$\frac{dU_m}{dt} = \int_{-1}^1 \sigma I d\Omega - c \sigma U_r, \quad (102b)$$

where the material energy density equation can be written equivalently as

$$\frac{1}{\beta} \frac{dU_r}{dt} = \int_{4\pi} \sigma I d\Omega - c \sigma U_r. \quad (102c)$$

We assume that the specific heat c_v is constant, implying that β is identically $\frac{4aT^3}{c_v}$, and that the opacity is inversely proportional to T^p :

$$\sigma(T) = \frac{\gamma}{T^p}, \quad 0 \leq p, \quad (103)$$

Next, we define the scaled time variable τ which is tied to an (arbitrary) equilibrium temperature T_0 :

$$\tau \equiv q \sigma_0 c t, \quad (104a)$$

where

$$q \equiv \frac{aT_0^3}{c_v}. \quad (104b)$$

Note that $\tau = 1$ can be interpreted as the mean free time between absorptions and reemissions at equilibrium. The scaled material energy density M is

$$M(\tau) = \frac{U_m[T(t)]}{U_m(T_0)} = \frac{T}{T_0}, \quad (104c)$$

and the scaled equilibrium radiative energy density R is

$$R(\tau) = \frac{U_r[T(t)]}{U_r(T_0)} = \frac{aT^4}{aT_0^4} = \left(\frac{T}{T_0} \right)^4. \quad (104d)$$

We also define a dimensionless scalar intensity:

$$\phi(\tau) = \frac{\int_{4\pi} I(\mathbf{\Omega}, t) d\Omega}{acT_0^4}. \quad (104e)$$

Noting that

$$\frac{d}{d\tau} = \frac{dt}{d\tau} \frac{d}{dt} = \frac{1}{q\sigma_0 c} \frac{d}{dt}, \quad (104f)$$

Eq. (102a) can now be rewritten in non-dimensionalized form by integrating over angle and using Eqs. (104)

$$\begin{aligned} q\sigma_0 \frac{d}{d\tau} (\phi acT_0^4) + \sigma(\phi acT_0^4) &= c\sigma(RaT_0^4), \\ q \frac{d\phi}{d\tau} + \frac{\sigma}{\sigma_0} \phi &= \frac{\sigma}{\sigma_0} R, \end{aligned} \quad (105)$$

while Eq. (102b) becomes:

$$\begin{aligned} \frac{aT_0^3}{c_v} \sigma_0 c \frac{d}{d\tau} (c_v T_0 M) + c\sigma(RaT_0^4) &= \sigma(\phi acT_0^4), \\ \frac{dM}{d\tau} + \frac{\sigma}{\sigma_0} R &= \frac{\sigma}{\sigma_0} \phi, \\ \frac{dM}{d\tau} + \frac{1}{M^p} R &= \frac{1}{M^p} \phi. \end{aligned} \quad (106)$$

or, equivalently [as for Eq. (102c)] by using $\frac{dR}{dM} = 4M^3$:

$$\frac{1}{4M^3} \frac{dR}{d\tau} + \frac{1}{M^p} R = \frac{1}{M^p} \phi. \quad (107)$$

To bring this together, the dimensionless form of the 0-D TRT equations applied to this nonlinear problem is:

$$q \frac{d\phi}{d\tau} + \frac{1}{M^p} \phi = \frac{1}{M^p} R, \quad (108a)$$

$$\frac{dM}{d\tau} + \frac{1}{M^p} R = \frac{1}{M^p} \phi, \quad (108b)$$

or

$$\frac{dR}{d\tau} + \frac{4}{M^{p-3}} R = \frac{4}{M^{p-3}} \phi. \quad (108c)$$

We next apply the IMC approximations to these equations (note that it is rigorously equivalent to begin from the IMC equations and non-dimensionalize). We begin by “freezing” the temperature-dependent coefficient data at the beginning of the time step τ_n :

$$q \frac{d\phi}{d\tau} + \frac{1}{M_n^p} \phi = \frac{1}{M_n^p} R, \quad (109a)$$

$$\frac{dM}{d\tau} + \frac{1}{M_n^p} R = \frac{1}{M_n^p} \phi, \quad (109b)$$

and now, *not* equivalently,

$$\frac{dR}{d\tau} + \frac{4}{M_n^{p-3}} R = \frac{4}{M_n^{p-3}} \phi. \quad (109c)$$

We next operate on Eq. (108c) with a time-averaging integral

$$\overline{(\cdot)} \equiv \frac{1}{\Delta_\tau} \int_{\tau_n}^{\tau_{n+1}} (\cdot) d\tau, \quad (110)$$

which produces

$$\frac{1}{\Delta_\tau} (R_{n+1} - R_n) + \frac{4}{M_n^{p-3}} \bar{R} = \frac{4}{M_n^{p-3}} \bar{\phi}, \quad (111)$$

and by approximating $\bar{R}$ as

$$\bar{R} = \alpha R_{n+1} + (1 - \alpha) R_n, \quad (112)$$

where α is user-defined such that $\frac{1}{2} \leq \alpha \leq 1$. Upon substitution of Eq. (112), Eq. (111) becomes:

$$\begin{aligned} \frac{1}{\alpha \Delta_\tau} (\bar{R} - R_n) + \frac{4}{M_n^{p-3}} \bar{R} &= \frac{4}{M_n^{p-3}} \bar{\phi}, \\ \bar{R} &= \frac{1}{1 + \frac{4\alpha \Delta_\tau}{M_n^{p-3}}} R_n + \frac{4\alpha \Delta_\tau / M_n^{p-3}}{1 + \frac{4\alpha \Delta_\tau}{M_n^{p-3}}} \phi_n. \end{aligned} \quad (113)$$

Defining the Fleck factor with the scaled data as

$$f_n \equiv \frac{1}{1 + \frac{4\alpha \Delta_\tau}{M_n^{p-3}}}, \quad (114)$$

the final approximation is to replace the time-averaged values in Eq. (113) with “instantaneous” values,

$$R(\tau) \approx \bar{R} = f_n R_n + (1 - f_n) \phi(\tau). \quad (115)$$

Introducing Eq. (115) into the radiation equation Eq. (108a), we obtain the dimensionless, radiation IMC equation (using $R_n = M_n^4$),

$$q \frac{d\phi}{d\tau} + \frac{f_n}{M_n^p} \phi = f_n M_n^{4-p} \phi, \quad (116a)$$

and, employing the same approximation in the material energy equation Eq. (108b), we obtain the dimensionless IMC equation for M ,

$$\frac{dM}{d\tau} + f_n M_n^{4-p} = \frac{f_n}{M_n^p} \phi. \quad (116b)$$

Adding these equations yields the conservation of energy equation in an infinite medium,

$$\frac{d}{d\tau} (q\phi + M) = 0, \quad (117)$$

which implies

$$q\phi(\tau) + M(\tau) = \text{constant} = q + 1, \quad (118)$$

where the final equality comes from the recognition that, at equilibrium, $\phi = M = 1$. Next, we use Eq. (118) to eliminate ϕ from Eq. (116b),

$$\frac{dM}{d\tau} + \frac{f_n}{qM_n^p} M = \frac{f_n}{qM_n^p} ((q + 1) - qM_n^4). \quad (119)$$

This first-order, ordinary differential equation for $M(\tau)$ may be solved with an integrating factor,

$$M(\tau) = M_n e^{-f_n(\tau - \tau_n)/qM_n^p} + ((q + 1) - qM_n^4) \left(1 - e^{-f_n(\tau - \tau_n)/qM_n^p}\right),$$

and thus the temperature at the end of the time step is

$$M_{n+1} = q + 1 - qM_n^4 + (M_n - q - 1 + qM_n^4) e^{-f_n \Delta\tau / qM_n^p}. \quad (120)$$

Eq. (120) is the *exact* solution of the 0-D, scaled IMC equations. However, it is not possible to simply rearrange this solution in the form $M_{n+1} = \rho M_n$ in order to determine the amplification factor ρ .

To obtain an amplification factor, we consider a first-order perturbation in M_n ,

$$M_n = 1 + \epsilon P_n,$$

and look for a recursion relationship for P_{n+1} . Noting that

$$M_n^k = (1 + \epsilon P_n)^k = 1 + k\epsilon P_n + O(\epsilon^2), \quad (121a)$$

$$f_n = \frac{1}{1 + 4\alpha\Delta\tau M_n^{3-p}} = f_0 + \epsilon P_n(3 - p)(1 - f_0) + O(\epsilon^2), \quad (121b)$$

$$M_n - q - 1 + qM_n^4 = 1 + \epsilon P_n - q - 1 + q(1 + 4\epsilon P_n) = \epsilon P_n(1 + 4q), \quad (121c)$$

Eq. (120) becomes, to first order,

$$\begin{aligned} 1 + \epsilon P_{n+1} &= (q + 1) - q(1 + 4\epsilon P_n) + \epsilon P_n(1 + 4q) e^{-f_0 \Delta\tau / q}, \\ P_{n+1} &= \left[(1 + 4q) e^{-f_0 \Delta\tau / q} - 4q \right] P_n. \end{aligned}$$

Finally, we identify the amplification factor,

$$\rho \equiv (1 + 4q) e^{-f_0 \Delta\tau / q} - 4q. \quad (122)$$

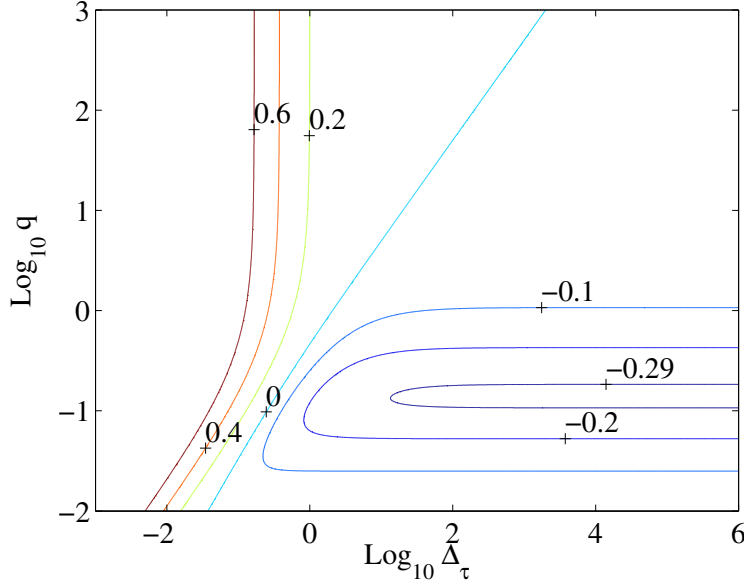


Figure 1: A contour plot of the 0-D amplification factor for various choices of $\Delta\tau, q$.

This expression is identical to that found in earlier work with the assumption $p = 3$ and noting that $f_0 = f$ in the original derivation (Wollaber, 2008; Wollaber & Larsen, 2011). In particular, this means that, no matter the value of p , the IMC equations are unconditionally stable, or $|\rho| \leq 1$. However, they permit damped temporal oscillations whenever $\rho < 0$, and ρ reaches an absolute minimum when $\Delta\tau \rightarrow \infty$ at -0.2984. A contour plot of ρ showing its sensitivity to time step size and q is provided as Figure 1.

This more general analysis has rigorously demonstrated that, to first order, the monomial order does not affect the stability characteristics of the IMC equations. In particular, this means that, no matter the value of p , the IMC equations are unconditionally stable, or $|\rho| \leq 1$ (Wollaber, 2008; Wollaber & Larsen, 2011). However, the IMC temperature solutions do admit damped temporal oscillations whenever $\rho < 0$, which can occur if the time step is chosen to be sufficiently large. Finally, having demonstrated this property for any inverse monomial power of T , we speculate that it is possible to demonstrate similar properties for polynomials and other functions well-approximated by combinations of monomials.

Fig. 2 depicts a time-dependent IMC solution containing damped oscillations for a problem with an inverse quintic temperature dependence ($p = 5$), $q = 0.14$, $\Delta\tau = 10$ and $M_0 = 0.9$, for which Eq. (122) predicts an amplification factor of about -0.2868. The linear stability analysis underpredicts the observed damped oscillations by a factor that is $O(\epsilon^2)$ for $\epsilon = 0.1$.

Although damped oscillations are clearly an undesirable characteristic of admissible solutions to the IMC or Taylor-expanded deterministic equations, they are not currently recognized as a major drawback. However, in coupling IMC with other physics such as non-equilibrium ion/electron temperatures, the behavior can become exacerbated (Wollaber, 2013a).

4.2.2. Maximum Principle Violations

Arguably, the most serious deficiency of the IMC equations is that they admit solutions that violate a *maximum principle* (Andreev et al., 1983; Larsen & Mercier, 1987). We remind the reader that, in this context, solutions that violate the maximum principle imply that interior temperatures can exceed boundary temperatures, even in the absence of internal sources. An illustration of this behavior for a scaled problem with a (dimensionless), isotropic, left boundary condition at 1 and an initial scaled temperature of 0.1 is depicted in Fig. 3 at time $\tau = 8$ with several time step sizes. Fig. 3 clearly illustrates how, by increasing time step sizes on a fixed spatial mesh, maximum principle violations are progressively exacerbated – the

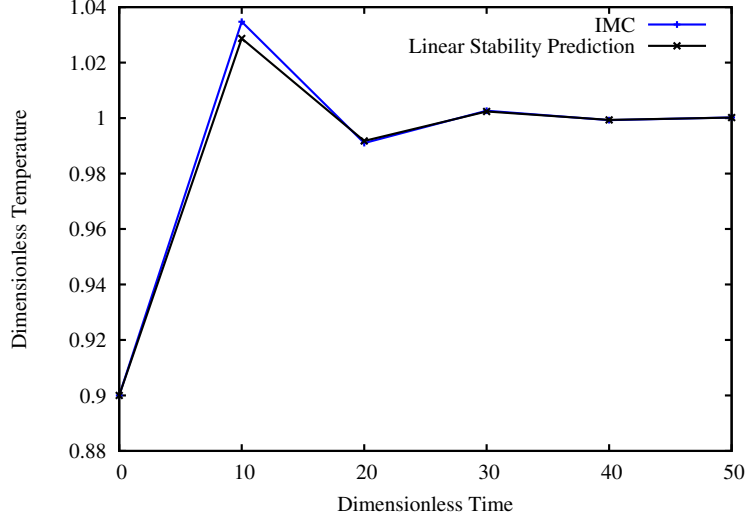


Figure 2: Damped temporal oscillations in IMC temperature profiles over five time steps for a near-equilibrium, 0-D problem with $\sigma \propto T^{-5}$.

thermal wave appears to move too slowly and “pile up”, well above the temperature of the boundary source. To demonstrate this, we briefly revisit a sufficient condition on the time step size first demonstrated by Larsen and Mercier for a gray problem assuming a constant (or time-explicit) specific heat. Denoting the maximum incident boundary temperature T_U , defining $B_U \equiv acT_U^4$ as the maximum permissible equilibrium radiative energy density, and assuming σ is frequency-independent, the gray IMC equations are

$$\frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I = \frac{1}{4\pi} \int_{4\pi} f_n \sigma_n I d\Omega + \frac{1}{4\pi} \sigma_n B_n, \quad (123a)$$

$$\frac{c_{v,n}}{\Delta t} (T_{n+1} - T_n) = f_n \sigma_n \int_{t_n}^{t_{n+1}} \left(\int_{4\pi} I d\Omega - B_n \right) dt. \quad (123b)$$

We presume that $T_n(x) \leq T_U$ by construction. Defining $\psi \equiv B_U - I$ and substituting ψ into Eq. (123a), we derive a transport equation for ψ ,

$$\frac{1}{c} \frac{\partial \psi}{\partial t} + \mathbf{\Omega} \cdot \nabla \psi + \sigma_n \psi = \frac{1}{4\pi} \int_{4\pi} f_n \sigma_n \psi d\Omega + \frac{1}{4\pi} \sigma_n (B_U - B_n). \quad (124)$$

Because $B_u \geq B_n$ by assumption, and because the remaining terms constitute a linear transport equation for ψ , we may conclude that $\psi \geq 0$, which also implies that $B_U \geq I$ (Case & Zweifel, 1967; Larsen & Mercier, 1987). We next consider Eq. (123b),

$$T_{n+1} = T_n + \frac{1}{c_{v,n}} f_n \sigma_n \int_{t_n}^{t_{n+1}} \left(\int_{4\pi} I d\Omega - B_n \right) dt, \quad (125)$$

and, since $B_U \geq I$,

$$T_{n+1} \leq T_n + \frac{\Delta t}{c_{v,n}} f_n \sigma_n (B_U - B_n), \quad (126)$$

so that, subtracting,

$$T_U - T_{n+1} \geq T_U - T_n - \frac{\Delta t}{c_{v,n}} f_n \sigma_n (B_U - B_n). \quad (127)$$

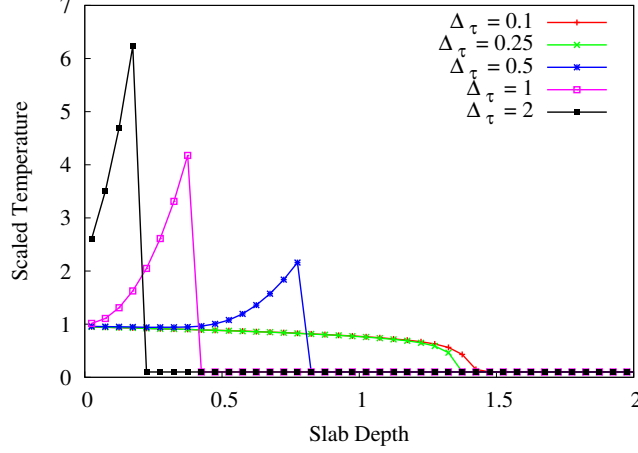


Figure 3: Temperature profiles at $\tau = 8$ for a Marshak wave problem in which the time step is varied using $\Delta\tau = 0.1, 0.25, 0.5, 1$, and 2 mean free times for emission.

We would like to demand that the left side of this expression ($T_u - T_{n+1}$) be positive. Because $T_U - T_n > 0$ and the rightmost term is positive, a sufficient condition for this to be true is to require that

$$T_U - T_n \geq \frac{\Delta t}{c_{v,n}} f_n \sigma_n (B_U - B_n). \quad (128)$$

Using the definitions of f_n , $\beta_n = 4aT_n^3/c_{v,n}$, and $B = acT^4$,

$$\begin{aligned} \frac{T_U - T_n}{T_U^4 - T_n^4} &\geq \frac{\sigma_n ac}{c_{v,n}} \frac{\Delta t}{1 + \alpha 4aT_n^3 c \sigma_n \Delta t / c_{v,n}}, \\ 1 &\geq \Delta t \left(\frac{\sigma_n ac}{c_{v,n}} \right) \left(\frac{T_U^4 - T_n^4}{T_U - T_n} - 4\alpha T_n^3 \right). \end{aligned} \quad (129)$$

Eq. (129), first derived by Larsen and Mercier, places a sufficient condition on the maximum allowable Δt such that $T_{n+1} \leq T_U$. It is a fundamental to the IMC equations – and the Taylor-expanded deterministic equations in Eqs. (45) – independent of any statistical or other numerical artifacts. However, in practice it was found that the limitation on Δt is too restrictive to be useful as a time step controller. This is because replacing I with B_U is too strong an assumption and, more importantly, that *spatial* temperature grids relax this restriction (Wollaber et al., 2013). Fig. 4 illustrates the dependence of $(\Delta x, \Delta t)$ on the onset of discrete maximum principle violations as predicted by theory provided in (Wollaber et al., 2013) for the first time step of a frequency-dependent problem from (Fleck & Cummings, 1971; Larsen & Mercier, 1987). For sufficiently small Δt and/or large Δx (the left and top parts of Fig. 4), the IMC solution does not violate the maximum principle. For large Δt and/or small Δx (the right and bottom parts of Fig. 4), the IMC solutions do violate the maximum principle.

We next consider the behavior of the IMC equations in a particular near-equilibrium condition.

4.2.3. Asymptotic Equilibrium Diffusion Limit

Even in the absence of spatial grid effects, the IMC equations do not preserve the “asymptotic equilibrium diffusion limit” due to the linearization error (Densmore & Larsen, 2004). This limiting regime occurs in many physically relevant problems and represents a system in which the intensity is well-approximated by a Planckian at temperature T , where T varies slowly in space and time with respect to photon mean-free-paths between collisions and absorption/re-emission (Larsen et al., 1983; Morel, 2000). More formally, we

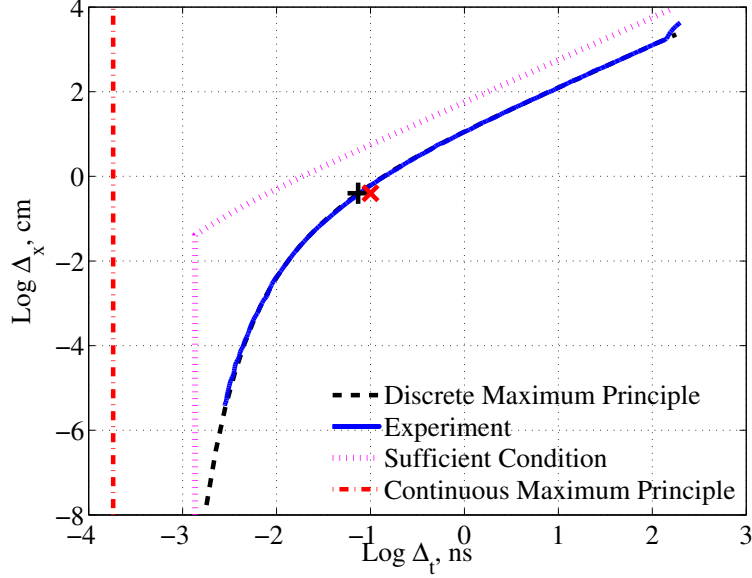


Figure 4: First incidences of theoretically predicted (dashed) and experimental (solid) maximum-principle violations for a frequency-dependent Marshak wave problem. The Δt from the continuous maximum principle (Larsen & Mercier, 1987) is the vertical line at the left. The black "+" indicates a problem simulated in (Larsen & Mercier, 1987) that did not violate the maximum principle despite using a much larger time step size than that indicated by the red line. The red "X" solution did violate the maximum principle.

introduce an asymptotic scaling with a small parameter ϵ into the gray TRT equations (we assume gray opacities for ease of presentation),

$$\frac{\epsilon}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \frac{\sigma}{\epsilon} I = \frac{\sigma}{\epsilon} \frac{acT^4}{4\pi}, \quad (130a)$$

$$\epsilon c_v \frac{\partial T}{\partial t} = \epsilon \frac{\partial U_m}{\partial t} = \int_{4\pi} \frac{\sigma}{\epsilon} \left(I - \frac{acT^4}{4\pi} \right) d\Omega. \quad (130b)$$

The limiting behavior of Eqs. (130) as $\epsilon \rightarrow 0$ is referred to as the asymptotic equilibrium diffusion limit. Solutions are formally obtained by assuming that the intensity and temperature may be expanded in a power series in ϵ ,

$$I = I^{(0)} + \epsilon I^{(1)} + \epsilon^2 I^{(2)} + O(\epsilon^3), \quad (131a)$$

$$T = T^{(0)} + \epsilon T^{(1)} + \epsilon^2 T^{(2)} + O(\epsilon^3), \quad (131b)$$

and equating like powers of ϵ . For instance, substituting Eqs. (131) into Eqs. (130) and equating the $O(1/\epsilon)$ terms produces the consistent set of equations

$$I^{(0)} = \frac{ac(T^{(0)})^4}{4\pi}, \quad (132a)$$

$$0 = \int_{4\pi} \left(I^{(0)} - \frac{ac(T^{(0)})^4}{4\pi} \right) d\Omega, \quad (132b)$$

implying that, to first order, the intensity is isotropic (and Planckian for frequency dependence). Equating the $O(1)$ terms yields another set of consistent equations

$$-\frac{1}{\sigma}\mathbf{\Omega} \cdot \nabla I^{(0)} = \left(I^{(1)} - 4 \frac{ac(T^{(0)})^3 T^{(1)}}{4\pi} \right), \quad (133a)$$

$$\int_{4\pi} \sigma \left(I^{(1)} - 4 \frac{ac(T^{(0)})^3 T^{(1)}}{4\pi} \right) = 0, \quad (133b)$$

which, by Eq. (132a) implies that

$$I^{(1)} = -\frac{1}{\sigma}\mathbf{\Omega} \cdot \nabla \frac{ac(T^{(0)})^4}{4\pi} + 4 \frac{ac(T^{(0)})^3 T^{(1)}}{4\pi}. \quad (134)$$

Equating the $O(\epsilon)$ terms produces

$$\frac{1}{c} \frac{\partial I^{(0)}}{\partial t} + \mathbf{\Omega} \cdot \nabla I^{(1)} + \sigma I^{(2)} = \frac{\sigma ac}{4\pi} (6T^{(0)}T^{(1)})^2 + 4(T^{(0)})^3 T^{(2)}, \quad (135a)$$

$$c_v \frac{\partial T^{(0)}}{\partial t} = \int_{4\pi} \sigma \left(I^{(2)} - \frac{ac}{4\pi} (6T^{(0)}T^{(1)})^2 + 4(T^{(0)})^3 T^{(2)} \right) d\Omega. \quad (135b)$$

Integrating Eq. (135a) over solid angle and substituting Eq. (132a) and Eq. (134),

$$\frac{\partial (aT^{(0)})^4}{\partial t} - \nabla \cdot \frac{1}{3\sigma} \nabla ac(T^{(0)})^4 = \int_{4\pi} \sigma \left(\frac{\sigma ac}{4\pi} (6T^{(0)}T^{(1)})^2 + 4(T^{(0)})^3 T^{(2)} - I^{(2)} \right) d\Omega. \quad (136)$$

Adding this result to Eq. (135b) provides a solvability condition for the leading order temperature $T^{(0)}$,

$$\frac{\partial (aT^{(0)})^4}{\partial t} + c_v \frac{\partial T^{(0)}}{\partial t} - \nabla \cdot \frac{1}{3\sigma} \nabla ac(T^{(0)})^4 = 0. \quad (137)$$

Eq. (137) is a nonlinear *equilibrium diffusion* equation for $T^{(0)}$ that is valid away from initial and boundary layers, where we recognize the diffusion coefficient as $D = 1/3\sigma$ (in a multifrequency setting, the reciprocal σ in D motivates the Rosseland opacity). Densmore and Larsen analyzed the IMC equations to determine whether they retain this limit under the same scaling. Following a procedure similar to the one above and incorporating an initial layer analysis first provided by Pomraning, they determined that, to leading order, the IMC equations satisfy

$$\frac{c_v}{\Delta t} (T_{n+1}^{(0)} - T_n^{(0)}) + \frac{1}{c\Delta t} (\phi_{n+1}^{(0)} - \phi_n^{(0)}) = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} \nabla \cdot \frac{1}{3\sigma_n^{(0)}} \nabla \phi^{(0)} dt, \quad (138a)$$

where $\phi^{(0)} = \int_{4\pi} I^{(0)}$ and $\phi^{(0)}$ also satisfies

$$\frac{1}{c} \frac{\partial \phi^{(0)}}{\partial t} - \nabla \cdot \frac{1}{3\sigma_n^{(0)}} \nabla \phi^{(0)} + \sigma_n^{(2)} \phi^{(0)} = \sigma_n^{(2)} ac(T_n^{(0)})^4. \quad (138b)$$

The pertinent result from this analysis is that Eqs. (138) do *not* constitute a valid time-discretized version of Eq. (137), as, although the leading order scalar intensity $\phi^{(0)}$ does satisfy a diffusion equation, it allows the radiation and material to stay out of equilibrium (compare with Eq. (132a)), and the source term contains only information from the beginning of the time step $T_n^{(0)}$. However, because this system does maintain a diffusive character, Densmore and Larsen note that the “method is capable of generating reasonable solutions, even when optically large time steps are used” (Densmore & Larsen, 2004). We also note that this analysis is independent of the spatial grid; poor choices of the spatially discretized temperature representation can further degrade the performance of IMC in the equilibrium diffusion limit.

4.3. Proposed Enhancements for Temporal Accuracy

In their seminal paper, Fleck and Cummings acknowledged that the time-explicit data in the IMC equations could lead to degraded accuracy and recommended linearly extrapolating temperatures from previous time steps (Fleck & Cummings, 1971). However, simple extrapolations on data that is already subject to statistical fluctuations can amplify noise at later time steps, so this practice is not widely used. There are, however, several approaches to enhancing the time-accuracy of the IMC equations.

A family of methods that enhance temporal accuracy through Fleck factor modifications is discussed in Section 2.4.2. For instance, the time-dependent Fleck factor in Eq. (53) or adaptive α in Eq. (57) restore the $O(\Delta t^2)$ accuracy for linear problems (Wollaber & Larsen, 2009b), although these remedies do not address the linearization errors. McClarren and Urbatsch’s “higher order integration” technique is designed to address maximum principle violations (McClarren & Urbatsch, 2009), and Wollaber’s synthetic Fleck factor attempts to combine the two techniques to address both short- and long-time step accuracy (Wollaber, 2011). Fleck factor modifications are appealing because they are simple to implement into existing IMC implementations and nominally require only cell-local data; however, they are not a panacea to the stability and maximum principle violations outlined above, as they cannot entirely circumvent linearization errors.

Cheatham developed a novel predictor-corrector algorithm (Cheatham et al., 2009) that is $O(\Delta t^2)$ so long as $\alpha = 1/2$ (or, perhaps, if the adaptive α is used). His technique proceeds by using a relatively small fraction of the IMC particles for a predictor step, followed by a larger fraction for the corrector step, and he explored the use of several means of opacity averaging. We note that, in general, it is undesirable to perform a full IMC time step simulation and “throw it away” a posteriori, as this requires duplicate storage of the beginning-of-time-step census and a significant amount of wasted computation, so “cheap” predictor steps are imperative. Alternatively, Wollaber has proposed hybrid deterministic-Monte Carlo methods for use as a predictor step (Wollaber & Larsen, 2009a; Wollaber & Warsa, 2013). In those methods, either an Eddington tensor or tensor diffusion coefficient is tallied during a time step, along with intensity-weighted opacity data, in order to facilitate the deterministic solution of a gray diffusion equation to predict the upcoming end-of-time-step temperature. The predicted temperature is then solely used to time-center the temperature-dependent problem data and is replaced by the ensuing IMC solution. As with Cheatham’s distributed particle approach, this approach is also $O(\Delta t)$ unless $\alpha = 1/2$ is employed. Both approaches require “extra” computations that may introduce their own statistical and/or algorithmic issues that balance their accuracy payoffs.

More closely related to temperature extrapolations, McClarren and Urbatsch extended the second-order backward difference formulation (BDF-2) to the IMC equations, outfitting the formulation with a lumping strategy for added temporal robustness and a means to avoid the calculation of opacity derivatives (Long et al., 2012; McClarren & Urbatsch, 2013, 2012). Although their lumping strategy gives up on theoretical second order time accuracy, in practice the linearization error is still first order, so the trade-off is aptly made. BDF-2 IMC reportedly reduces maximum principle violations and damped temporal oscillations under certain circumstances, and it has been successfully demonstrated in an ICF problem (McClarren & Urbatsch, 2013).

Gentile has proposed an extension of the IMC equations that incorporates temperature-derivatives of the opacities (Gentile, 2007, 2011). Opacities can be strongly temperature-dependent, implying that their time-explicit approximation can be a leading source of error (Cheatham, 2010; Gentile, 2011), so accounting for their variation within the time step can significantly reduce the error. He derived a technique that essentially updates the Fleck factor to account for the opacity derivative, implying that existing IMC implementations can immediately take advantage of this method (Wollaeger et al., 2013). Gentile also posed and solved a simple, but challenging problem for IMC involving $\sigma \propto T^{-5}$ using the approach (this is a particularly challenging problem because the emission term, $\sigma a T^4$, becomes small as T increases, so overheating errors can persist for many time steps). It appears that the only drawback of this approach is the difficulty in computing temperature derivatives from the opacity data; the method appears to successfully enhance accuracy and stability for temperature-dependent opacity data for both short and large time steps.

However, the only likely “true” remedy to prevent maximum principle violations is to abandon IMC in favor of a truly time-implicit method (Larsen et al., 2013). The price of these methods, however is

that they typically necessitate an iterative nonlinear solve. The most developed of these is the “Symbolic” Implicit Monte Carlo approach, which chooses the latter (Brooks, 1989; McKinley et al., 2003; N’kaoua, 1991; N’Kaoua, 1991). In that method, a collision probability matrix is created via Monte Carlo particle tracking to determine the cell-to-cell radiation coupling in the problem. The matrix is then inverted to determine the end-of-time-step temperatures. Special treatments are necessary for very thick, absorbing cells (such as an equilibrium diffusion approximation) that would otherwise generate very little coupling data, but the method is progressing under active development.

Park et al. have proposed a “High-order, low order” (HOLO) approach that, through a “consistency term”, moves the nonlinear convergence of the time-implicit temperature to a moment-reduced low-order system (Park et al., 2013a,b,c, 2014). This approach is similar enough to IMC that existing IMC implementations can be boot-strapped for algorithm deployment, although they also can require special treatments in highly absorbing regions to avoid seeding noise into the consistency terms. Willert and Park have extended the results to calculate a correction to the time-implicit high-order equations (Willert & Park, 2014), while Bolding et. al are pursuing a related, alternative path for “exponentially convergent” Monte Carlo (Bolding et al., 2015). All of these approaches require the external solution of a finite-volume or finite-element, reduced order system. “High-order, low order” approaches are an active area of promising research that may lead to the deprecation of the IMC method.

Long recently proposed an “iterative thermal emission” (ITE) approach, primarily to address maximum principle violations (Long et al., 2014; Long, 2012). The method uses the Rosenbrock semi-implicit scheme (Rosenbrock, 1963) as a sub-cycling approach to iterate the end-of-time-step temperature, again, primarily to circumvent maximum principle violations otherwise produced by the IMC equations. It can also be implemented through modifications to the Fleck factor during each subcycle and by partitioning the particle energy in each subcycle. The method was found to produce more accurate results on nonlinear problems subject to maximum principle violations without requiring a global matrix inversion. However, Long found that the method’s variation in the Fleck factor leads to higher statistical errors as compared with the IMC equations.

Gentile and Yee have proposed an Iterative Implicit Monte Carlo approach, similar in spirit to Long’s work, but fully implicit (Gentile & Yee, 2015). It also avoids the construction and inversion of a spatially global matrix as in SIMC (Brooks, 1989). However, the method requires more development to accelerate the iterative procedure, which can become arbitrarily slow for thick problems.

Although time-accuracy issues have attracted much attention, the IMC method cannot be implemented without the imposition of a temperature field that is typically represented in a spatially discrete form (this is in contrast to many other neutral-particle Monte Carlo problems that can be performed on mesh-free, combinatorial geometry representations to theoretically obtain “exact” spatial problem representations). Typically, other data (β, σ) are also represented on the discrete temperature field. These lead to spatial grid errors in the numerical solutions of the IMC equations.

5. Spatial Grid Effects

The imposition of a spatial grid in the IMC equations leads to an inaccuracy now referred to as “teleportation error” (Cheatham, 2010; Densmore, 2011; Irvine, 2013; Long, 2012; McKinley et al., 2003). This effect occurs when Monte Carlo particles are absorbed in one time step and non-physically re-emitted ahead of a wavefront during the next time step, artificially increasing the wave speed in problems that are emission-dominated. The effect is exacerbated by increasing the spatial grid size and/or by decreasing the time step size (because larger time step sizes increase the proportion of effective scattering over effective absorption, less energy can be re-emitted ahead of the wavefront (McKinley et al., 2003)). This is in contrast to maximum principle violations (overheating) that are exacerbated by decreasing the spatial grid size and/or increasing the time step size (Wollaber et al., 2013). “Teleportation error” has also been used to refer to errors that arise by particular choices of the spatial representation of the T^4 emission term in the equilibrium diffusion limit, although this apparent teleportation (increased wave speed) technically arises due to other considerations, discussed next (Smedley-Stevenson & McClarren, 2013).

5.1. Diffusion Limit Accuracy

It has already been recognized for spatially discrete thermal radiative transfer systems (in particular, discrete ordinates) that, to maintain accuracy in the asymptotic diffusion limit, the discrete transport system should limit to a meaningful discretization of the continuous diffusion equation (Adams & Nowak, 1998; Morel et al., 1996). Seemingly benign choices of spatial discretizations can lead to discrete diffusion equations that do not contain any opacity dependence to first order, for instance. Densmore has provided an initial analysis of the interaction of a spatially continuous radiation field (such as occurs in a Monte Carlo solution of the IMC equations) with discrete histogram and linear subcell temperature representations (Densmore, 2011). In that work, Densmore demonstrated that histogram temperature representations only produce a valid discrete diffusion equation for Fleck factors of $O(\epsilon^2)$, implying that optically large time steps are a prerequisite for accurate spatial solutions in IMC calculations. This spatial inaccuracy arises from mismatched, discontinuous cell-boundary and cell-interior intensities that arise for Fleck factors larger than $O(\epsilon^2)$. Because the Fleck factor is $O(1)$ as $\Delta t \rightarrow 0$, histogram temperatures cannot maintain accuracy under time step resolution. This behavior was observed empirically by Fleck and Canfield, who recommended using a linear subcell shape in T^4 (Fleck & Canfield, 1984). In 1D, this can be written as

$$T^4(x) = T_j^4 [1 + m_j(x - x_j)] , \quad (139)$$

where x_j is the center of cell j and m_j is an estimate of the normalized cell slope,

$$m_j \approx \frac{1}{T_j^4} \left. \frac{dT^4}{dx} \right|_j . \quad (140)$$

Typically, the slope values have been reconstructed using an ad-hoc interpolatory scheme based on temperatures in the surrounding grid locations. This sub-cell spatial representation of the emission source is often referred to as *tilt* or *source tilting*. Although it was recognized relatively early that source tilting is necessary to maintain spatial accuracy in the diffusion limit, developing an optimal prescription for sub-cell spatial emission is an ongoing area of research.

Densmore demonstrated that for reasonable estimates of m_j (for example, m_j could be obtained from the continuous diffusion equation), spatial accuracy in the diffusion limit can be maintained for Fleck factors of $O(1)$ through $O(\epsilon^2)$. However, for practical radiation-hydrodynamics problems, T^4 may not satisfy an underlying diffusion equation, and an m_j calculated from an ad-hoc interpolation scheme can again permit $O(1)$ discontinuities in the $T^4(x)$ emission source. Densmore’s analysis successfully predicts that, in such situations, large time steps (small Fleck factors) ameliorate spatial inaccuracies, and it implies that linear subcell representations of $T^4(x)$ are a necessary condition for spatial accuracy under arbitrary time step resolution. Under time step resolution, though, the errors that are introduced by the discontinuities are exacerbated. These issues are avoided to some extent in deterministic transport algorithms by explicitly representing the subcell shape of both the radiation field and the emission source via finite elements.

Smedley-Stevenson and McClarren provide a compelling argument that, for spatio-temporal accuracy in the equilibrium diffusion limit, one should prefer a *continuous* representation of $T^4(x)$ (Smedley-Stevenson & McClarren, 2013). By making assumptions that amount to an asymptotic equilibrium diffusion limit analysis, they find that the discrete diffusion equation that results from a linearly discontinuous emission profile contains “jump” source terms on cell faces with magnitudes proportional to the discontinuity. Because these jump emission terms are independent of problem data (such as opacities), and because they are not balanced by corresponding absorptions, they non-physically impose energy transfer between cell faces, hastening the transmission of a thermal wave. For the IMC equations, these anomalous source terms are reduced in magnitude for optically large time steps (small Fleck factors), in agreement with Densmore’s findings for histogram emission sources with $O(\epsilon^2)$ Fleck factors. Although Smedley-Stevenson and McClarren admit that it may be possible to construct linear discontinuous emission profiles whose discontinuities vanish sufficiently rapidly ($O(\epsilon)$ or better) as the thick diffusion limit is approached, they recommend a continuous sub-cell temperature representation containing two linear profiles that pass through the average temperature at the cell-centroid to entirely prohibit the anomalous source terms. They demonstrate excellent accuracy for

a Marshak wave problem with various spatial resolutions. Their particular reconstruction does not preserve the cell-averaged emission strength, and it is not yet clear what (if any) the consequences are for spatial accuracy away from the equilibrium diffusion limit.

Wollaeger et al., Clouet and Samba, and Brooks et al. have explored using a linear-discontinuous, finite element trial space to represent temperature (Brooks et al., 2006; Clouët & Samba, 2004; Wollaeger et al., 2014). The Brooks and Clouet references considered the SIMC method instead of IMC. Because the SIMC system of equations produces a time-implicit temperature profile, it has the advantage of maintaining the diffusion limit. In particular, SIMC’s implicitness allows both the radiation and temperature profiles to evolve together, ensuring that $I = B + O(\epsilon)$ at both the beginning and the end of the time-cycle. However, SIMC does not contain a Fleck factor to reduce the magnitude of any temperature discontinuities between cells, so its spatial accuracy is far more sensitive to the choice of sub-cell temperature profiles than is IMC. Wollaeger et al. reported enhanced accuracy at a given spatial resolution using linear-discontinuous finite elements for emission and absorption as compared with an interpolatory scheme (Wollaeger et al., 2014). He also employed sub-cell, spatially dependent opacity treatments that can have counterintuitive effects on spatial accuracy as they interplay with different “tilt” treatments (for instance, using a more accurate, sub-cell opacity in conjunction with a “standard” interpolatory tilt can decrease overall accuracy). IMC with a linear discontinuous, finite element discretization of the temperature was also shown to limit to a valid, cell-edge diffusion equation that contains $O(\epsilon)$ jump terms (Wollaeger et al., (submitted)). On the whole, current literature on “tilt” strongly suggests that subcell emission profiles must be at least linear and should contain asymptotically small discontinuities as they approach the equilibrium diffusion limit to maintain accuracy.

5.2. Teleportation Error

IMC simulations also suffer a tightly related spatial inaccuracy commonly referred to as *teleportation error*, which we define here as a significant, cycle-to-cycle difference between the (subcell) spatial profiles of the absorbed MC particle locations and their subsequent emission locations. McKinley et al. first introduced the term by highlighting how the behavior manifests in Marshak-wave problems, with the following description (McKinley et al., 2003).

Photon teleportation is an error that arises when the absorption mean free path is small compared to the size of a zone, and the time step size is also small (in the case for IMC). As photons stream through a zone they are attenuated by absorption. If the opacity is high, photons entering one side of a zone are completely absorbed very near the boundary. The scoring of energy deposition, however, is done as if the absorption had taken place evenly across the entire zone. On the next time step, the corresponding spontaneous emission is handled assuming this uniformity in the zone. In effect, spontaneous emission occurs too early on the far side of the zone and energy is transported across the problem too quickly. This error is aggravated in IMC if the time step size is reduced without a coordinated adjustment to the zone size.

Fig. 5 provides a graphical depiction of this discrepancy between the absorption and emission locations for a thick, diffusive Marshak wave problem ($\sigma = 300T^{-3} \text{ cm}^{-1}$, $T_0 = 1.0 \times 10^{-6} \text{ keV}$, left boundary at 1 keV, $\Delta t = 10 \text{ ps}$, $\Delta x = 0.005 \text{ cm}$, $c_v = 0.3 \text{ GJ/cc-keV}^4$). To generate these plots, we used 9 equally spaced subcell bins per spatial cell in Jayenne software (Wollaber, 2013b). The “normalized temperature” was calculated by normalizing the fourth root of the average energy deposition in the left-most cell to unity. The absorption profile is plotted from the previous time cycle, and the emission profile is plotted in the current time cycle. The jaggedness in these profiles arises from the discontinuities at cell interfaces in this particular tilt scheme and from the discontinuous absorption opacities between cells. At the left side of two cells near the edge of the wavefront, there is substantially more absorption than emission in all four subfigures – particles are frequently absorbed just as they enter a cell. In the right parts of these two leading cells, the emission is typically larger than the absorption, which is the “teleportation” effect.

Because this behavior can manifest in exactly the same way as in the diffusion limit inaccuracy – by artificially accelerating a wavefront – the term “teleportation error” has also been used in the literature

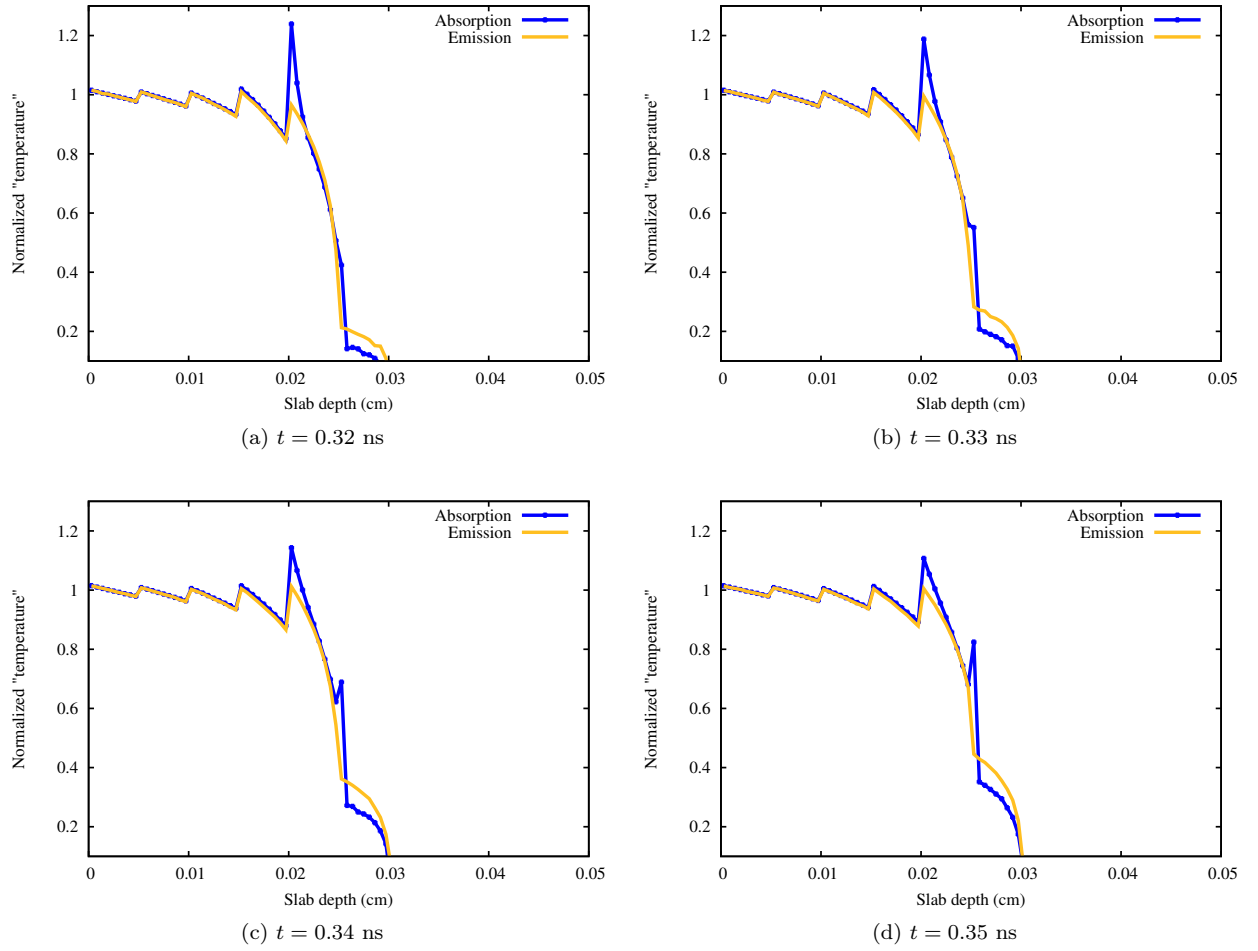


Figure 5: An illustration of “tilt” and the evolution of teleportation error over four time cycles. Note the prevalence of absorption in the left sides of the two leading cells near the wave-front and the enhanced emission over absorption in the right sides of those cells.

to describe inaccuracies in that limit. However, teleportation error may occur away from the equilibrium diffusion limit when subcell spatial gradients in emission locations cannot resolve the emission profile. We next discuss research to enhance the subcell absorption/emission accuracy to address teleportation errors without regard to that limit.

Methods improvements targeted at reducing teleportation error typically involve “scoring” additional spatial moment information, such as a slope. Wollaeger et al. have reported enhanced accuracy by straightforwardly applying a linear, discontinuous finite element method (DFEM) to the temperature profile and ensuring that particles absorptions are correctly integrated against the DFEM basis functions (Wollaeger et al., 2014). Wollaeger also explored the use of subcell, spatially variable opacities, buying further accuracy at the price of a more difficult calculation of the particle collision distance (Brown & Martin, 2003). Cheatham employed “functional expansion tallies” (FETs) using entropy-based Legendre expansions – that ensure positivity – and also reported significant accuracy enhancements. (Cheatham, 2010; Cheatham et al., 2010b). Holgado et al. are researching the use of “kernel density estimators” (KDEs), which are a mesh-free technique for tallying particle absorptions, although they have not yet explored sampling from them for particle emission locations (Holgado et al., 2015).

To reduce teleportation error without relying on an interpolation of the temperature grid or functional form of the previous absorption locations, Irvine et al. suggested storing particle absorption locations and ensuring that particles in the ensuing cycle sample their emission locations among the absorption locations from the previous cycle (Irvine, 2013; Irvine et al., 2015). In the KULL ICF code (Rathkopf et al., 2000), Irvine reported enhanced accuracy for coarser spatial zoning at a relatively modest (25%) increase in memory usage for 3D problems and about a 13% decrease in compute time. One difficulty with this approach is that emission locations are only selected from absorption locations in the cycle immediately preceding the current cycle; absorption locations from two or more cycles preceding the current cycle are discarded, even if, due to the choice of time step size, these ought to be physically valid emission locations. In particular, for asymptotically small time steps with respect to an average time between absorption and emission, vanishingly few absorptions occur in a given cell. Therefore, for this method to be robust, some further development is required to provide a time-window of absorption locations or to smoothly transition back to sampling a “tilt” when emission locations are unavailable.

6. Acceleration Methods and Algorithms

Unfortunately, a straightforward Monte Carlo implementation of the IMC method can quickly lead to computational difficulties such as prohibitively long runtimes to obtain solutions with an acceptable level of statistical noise. Reducing the number of particles to obtain reasonable runtimes can then lead to a situation containing all of the noise of a Monte Carlo simulation with the discretization errors from a deterministic simulation. Fortunately, there are several ways of accelerating the computation and/or reducing the statistical variances in the solution.

6.1. Parallel Computation Considerations

The most straightforward means of reducing the time-to-solution is to recognize the inherent parallelism in Monte Carlo simulations – distributing the particles among many distributed or shared memory processing elements (PEs) and modeling them simultaneously. Unfortunately, the memory requirements due to representing temperatures and disparate multigroup opacities on individual mesh cells can quickly become prohibitive, so it becomes necessary to employ spatial domain decomposition in addition to or instead of particle replication (Brunner & Brantley, 2009; Gentile, 2009; O’Brien et al., 2013). Maintaining reasonable load balance in spatially domain decomposed Monte Carlo simulations is a challenging requirement that remains an area of active research.

Additionally, it is desirable to enable reproducible solutions regardless of the parallel topology to ease the burden of debugging. Typically this is done by associating unique random number streams with individual particles (Brunner & Brantley, 2009; Brunner et al., 2006; Gentile et al., 2006; Wollaber, 2013b). For the ambitious developer, it is also possible to employ integer math operations to prevent the accumulation of roundoff from causing divergences in particle histories (Cleveland et al., 2013).

Parallel computation on heterogeneous architectures with shared-memory parallelism, such as the IBM Cell processor, GPUs, Intel MICs, etc., can also reduce time to solution, although these typically still require considerable developer effort to effectively map the stochastic algorithm and indirect data access requirements onto specialized hardware (Fang & Boas, 2009; Heymann & Siebenmorgen, 2012; Kelley et al., 2010; Xu et al., 2014).

However, parallelism alone is not and has not been the most effective means of reducing times-to-solution in IMC simulations.

6.2. Approximate Acceleration Methods

For IMC simulations that contain mesh cells near the equilibrium diffusion limit and that attempt to employ $O(1)$ time step sizes with respect to $O(1)$ changes in the material temperature, the Fleck factor becomes $O(\epsilon^2)$ (Densmore, 2011). Hence, the scattering ratio approaches unity in cells that are very optically thick $O(1/\epsilon)$, and Monte Carlo simulations become exceedingly long as each particle can scatter an arbitrarily large number of times in these cells while at the same time depositing very little energy. This is also true in thick, scattering-dominated problems, whether or not the cell is near the equilibrium diffusion limit. It has been a source of endless frustration for Monte Carlo developers that, in these regimes for which deterministic diffusion solutions are highly accurate and much more efficient, the presumably more accurate transport solution can take arbitrarily long to simulate.

Fleck and Canfield recognized this problem relatively early and recommended a “diffusion random walk” procedure to significantly speed up the Monte Carlo procedure at little reduction in solution accuracy (Cleveland, 2010; Cooper, 2000; Fleck & Canfield, 1984; Giorla & Sentis, 1987). The essential idea is to recognize that IMC particles in thick, highly scattering cells satisfy a diffusion equation over a sphere inscribed within the cell and centered on the particle location (and that we can pre-tabulate the known analytical solution of the diffusion equation over a sphere) (Fleck & Canfield, 1984). In this way, the particles can immediately be moved from the cell center to a sampled point on the sphere, potentially saving a great many computational scatter events. Particles are only selected for this “diffusion random walk” treatment if they meet certain criteria; for example, they must be isotropically distributed (so, they must have just emerged from a volume emission or scattered at least once), the enclosing sphere must have a radius of at least one Rosseland mean-free-path (the inverse of the Rosseland opacity), and the next scattering event must have been chosen to occur within the sphere. The advantage of this approach is that the technique may be attempted after every isotropic emission or scattering event, regardless of the particle’s location in the problem geometry, and each time that it is used, it replaces many scattering events with a single step. The disadvantage is that, for small cells and time-step sizes, it is much harder to satisfy the requirements to initiate the technique, so the computational returns diminish under refinement.

More recently, “discrete diffusion Monte Carlo” (DDMC) or “Implicit Monte Carlo Diffusion” (IMD) approaches have emerged that entirely replace transport zones with diffusive zones⁶ (Cleveland & Gentile, 2014; Cleveland et al., 2010; Densmore et al., 2012; Gentile, 2001; Wollaeger et al., 2013). These techniques all pre-identify diffusive problem regions before the calculation proceeds according to a set of diffusion criteria (Cleveland & Gentile, 2014) and then *replace* the IMC system in those regions with a spatially discretized diffusion system, usually on a per-frequency-group basis for multigroup problems. The discretized diffusion equation is provided with a Monte Carlo interpretation either by considering a stochastic matrix inversion of the diffusion equation (Cleveland et al., 2010; Gentile, 2001) or by reinterpreting the diffusion equation coefficients using probabilities (Densmore et al., 2012; Wollaeger et al., 2013). In DDMC, IMC particles traveling through transport zones that enter a DDMC zone are converted into DDMC particles such that they lose their exact position and angular direction within the DDMC zone. These DDMC particles do not scatter; instead their fates are to leak through a DDMC face, go to census, become absorbed, or (in multigroup problems) convert to an IMC particle in a thinner opacity group. The bulk removal of scattering events provides significant computational savings, sometimes by orders of magnitude. An additional advantage

⁶The primary difference between modern implementations of DDMC and IMD is that IMD contains a time-discretization in addition to the spatial discretization (Cleveland & Gentile, 2014).

is that, depending on the details of the spatial diffusion discretization and the “tilt”, spatial accuracy can actually improve in the diffusion limit. The disadvantage of these approaches is that there is computation and memory overhead that must be overcome in pre-selecting the diffusion regions, particularly, calculating face-centered opacities.

6.3. Variance Reduction and Population Control

We have assumed that the Monte Carlo calculation employs a constant energy-weight procedure. In this procedure, when a particle undergoes a collision, it is either absorbed in the material with probability f_n – at which point its entire energy-weight is deposited into the material – or scattered with probability $(1 - f_n)$ (in the absence of physical scattering). Therefore, the probability of a particle surviving after i collisions in a material with Fleck factor f_n is $(1 - f_n)^i$. Since particles usually undergo many collisions as they traverse the problem domain from highly populated to lightly populated regions, the probability of finding any given particle in a lightly populated region can become geometrically small. If obtaining good statistics in these regions is important, one must either geometrically increase the number of particles simulated, or modify the Monte Carlo procedure in such a way that (i) the probability of particle survival is increased in these regions and (ii) the mean solutions do not become biased. In general, techniques that attempt to modify the Monte Carlo procedure in order to increase the confidence in the solution are referred to as *variance reduction* methods. In this section, we briefly introduce four of the most widely-used variance reduction techniques and discuss how they affect the implementation of Eqs. (67). This section is *not* intended to fully explicate these methods; the interested reader should refer to one of several textbooks that discuss these topics, e.g. (Brown & Sutton, 1996; Hammersly & Handscomb, 1964; Kalos & Whitlock, 1986; Spanier & Gelbard, 1969).

6.3.1. Absorption Weighting

The first technique is one of the most widely-used methods in global Monte Carlo particle transport problems, and has been referred to alternately as *survival biasing*, *implicit capture*, or *absorption weighting*. In this technique, instead of depositing all of the particle’s energy-weight with probability f_n upon a collision event, we deposit the fraction f_n of its energy-weight into the material and continue to follow the fraction $(1 - f_n)$ of its energy-weight in the radiation. The weight update kernel for this procedure is therefore

$$p_{\text{a.w.}}(w' \rightarrow w) = \delta[w - (1 - f_n)w'], \quad (141)$$

where “a.w.” stands for absorption weighting. If Eq. (141) is used instead of Eq. (67c) in Eq. (67a) (or Eq. (94a)), then particle histories can no longer be terminated due to collision events. This means that each particle history will become longer, increasing the overall cost of simulating each particle. Also, instead of finding particles with weight w_0 in lightly populated regions with probability $(1 - f_n)^i$, one may expect to find particles in these regions with *weights* given by $(1 - f_n)^i w_0$, on average, where i is the number of collisions and w_0 is the initial energy-weight.

Although absorption weighting successfully provides a means for Monte Carlo particles to move from highly populated to lightly populated regions of the problem, it comes at the cost of longer particle histories. Also, because each particle can be expected to undergo a variable number of collisions i before it arrives in a lightly populated region of the problem, the variance in its weight can become large in these regions. The upshot is that, although the statistics in the lightly populated regions are improved, the cost of the calculation is increased in the highly populated regions (where the solution is already satisfactory) and there is room for improvement in the lightly populated regions. Additionally, absorption weighting allows particles to have arbitrarily small weights and removes a mechanism for “killing” particles, increasing their computational lifetimes. In practice, this can be remedied by switching back to “analog” tracking (sampling a distance-to-collision and killing absorbed particles) whenever the particle energy-weight has fallen below a particular cutoff. Although tempting, we do not recommend simply killing these low-weight particles and dumping their energies into the material, as that practice does introduce a bias.

6.3.2. Continuous Energy Deposition

A widely used improvement to absorption weighting for IMC problems is *continuous energy deposition* (Fleck & Cummings, 1971). In this technique, one entirely removes absorption “events” from the tracking procedure and modifies the distance-to-collision with a distance-to-scattering event; in Eq. (83d), one replaces σ with the smaller effective scattering opacity, $(1 - f)\sigma$ (if the problem contains physical scattering, that is added to the effective scattering). The particle then exponentially attenuates its energy-weight throughout its flight,

$$w(s) = w_0 e^{-f_n \sigma_n s}, \quad (142)$$

where w_0 is the particle’s initial energy-weight, $f_n \sigma_n$ is the effective absorption opacity, and s is the distance traveled. Eq. (67a) must therefore be modified to account for weight changes along a characteristic path. By the chain rule, the streaming term (in Cartesian coordinates) is

$$\frac{d\tilde{I}}{ds} = \frac{dt}{ds} \frac{\partial \tilde{I}}{\partial t} + \frac{dx}{ds} \frac{\partial \tilde{I}}{\partial x} + \frac{dy}{ds} \frac{\partial \tilde{I}}{\partial y} + \frac{dz}{ds} \frac{\partial \tilde{I}}{\partial z} + \frac{dw}{ds} \frac{\partial \tilde{I}}{\partial w}, \quad (143)$$

which, noting that $\mathbf{x} = \mathbf{x}_0 + \boldsymbol{\Omega}s$, $s = c(t - t_0)$, and, using Eq. (142), is

$$\begin{aligned} \frac{d\tilde{I}}{ds} &= \frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \tilde{I} - w_0 f_n \sigma_n e^{-f_n \sigma_n s} \frac{\partial \tilde{I}}{\partial w}, \\ &= \frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \tilde{I} - f_n \sigma_n w \frac{\partial \tilde{I}}{\partial w}. \end{aligned} \quad (144)$$

Additionally, because we have removed absorption events, the collision term contains only scattering events, making it $(1 - f_n)\sigma_n \tilde{I}$ instead of $\sigma_n \tilde{I}$. Finally, there are no weight changes upon particle scatter, so Eq. (67a) with continuous energy deposition becomes

$$\begin{aligned} \frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \boldsymbol{\Omega} \cdot \nabla \tilde{I} - f_n \sigma_n w \frac{\partial \tilde{I}}{\partial w} + (1 - f_n)\sigma_n \tilde{I} &= \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \iiint \delta(w' - w) (1 - f_n)\sigma_n \tilde{I} d\Omega' d\nu' dw' \\ &+ f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{cU_{r,n}}{4\pi} \delta(w - w_0) + (1 - f_n) \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{Q_m}{4\pi} \delta(w - w_0) + \frac{Q_r}{4\pi} \delta(w - w_0). \end{aligned} \quad (145)$$

Eq. (145) correctly reduces to Eq. (24a) by operating with $\int_0^\infty (\cdot) dw$ and using Eq. (66),

$$\begin{aligned} \frac{1}{c} \frac{\partial I}{\partial t} + \boldsymbol{\Omega} \cdot \nabla I - \int_0^\infty f_n \sigma_n w \frac{\partial \tilde{I}}{\partial w} dw + (1 - f_n)\sigma_n I &= \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \iint (1 - f_n)\sigma_n I d\Omega' d\nu' \\ &+ f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{cU_{r,n}}{4\pi} + (1 - f_n) \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{Q_m}{4\pi} + \frac{Q_r}{4\pi}. \end{aligned} \quad (146)$$

The differential weight term, upon integration, is

$$- \int_0^\infty f_n \sigma_n w \frac{\partial \tilde{I}}{\partial w} dw = f_n \sigma_n I - f_n \sigma_n w \tilde{I} \Big|_0^\infty = f_n \sigma_n I, \quad (147)$$

provided that $w\tilde{I}(w) \rightarrow 0$ as $w \rightarrow \infty$, which is a reasonable demand on any practical Monte Carlo implementation. We therefore recover the correct mean collision term in Eq. (24a) ($\sigma_n I$) by noting that substituting the result of Eq. (147) into Eq. (146) recovers the collision term, $f_n \sigma_n I + (1 - f_n)\sigma_n I = \sigma_n I$. Continuous energy deposition is therefore an unbiased procedure in IMC, as it reproduces the correct mean behavior. As with absorption weighting, this process also creates particles with arbitrarily small weights; this can again be remedied by switching to analog tracking.

Although absorption weighting and continuous energy deposition successfully provide a means for Monte Carlo particles to move from highly populated to lightly populated regions of the problem, it comes at the cost of longer particle histories. Also, because each particle can be expected to undergo a variable number of

scattering collisions i before it arrives in a lightly populated region of the problem, the variance in its weight can become large in these regions. The upshot is that, although the statistics in the lightly populated regions are improved, the cost of the calculation is increased in the highly populated regions (where the solution is already satisfactory) and there is room for improvement in the lightly populated regions. The next three variance reduction methods attempt to remedy these issues by restricting the particle weights in such a way that the efficiency of the calculation is improved and their variance is diminished.

6.3.3. Controlling Particle Weights

If the correct average particle weight could be obtained through some procedure (an area of active research), then one would be able to identify particles whose weights are so small as to not significantly contribute to the solution. The calculation time could be decreased if one could effectively “combine” these low-energy-weight particles into a single, average-energy-weight particle. The technique known as *Russian roulette* does something like this.

In the traditional version of Russian roulette, we define w_{low} to be a lower energy-weight cutoff and w_{avg} to be the desired average energy-weight. Then if a particle with energy-weight w undergoes a collision and $w < w_{\text{low}}$, with probability $p = w/w_{\text{avg}}$ (a relatively low probability) we increase the particle’s energy-weight to w_{avg} , and with probability $(1 - p)$ (a relatively high probability), we terminate the particle history. This technique does not conserve energy or momentum since, with probability p , the energy $w_{\text{avg}} - w$ is created “from nowhere”, and with probability $(1 - p)$, the energy w is lost. In previous work, we considered accounting for per-particle energy conservation by storing rouletted particle energy in a cell-dependent “bank” and only permitting particle creation if sufficient energy existed in the bank (Wollaber, 2008; Wollaber & Larsen, 2009a). Any remaining energy in the bank could then be added to the particle census to ensure conservation, however, this appeared to lead to another mechanism of teleportation error. A bias-free, energy-conserving, but less efficient alternative is to mark the particle for “analog” tracking and allow it to die upon its next absorption.

We next turn our attention to particles whose energy-weights are much higher than average. Such particles should occur relatively rarely, but they have a large effect on the estimate of the mean and, and, therefore, also drive up the variance of that estimate. If a reasonable approximation of the mean energy-weight of particles in a particular region of phase space is known, then it is possible to distribute a high energy-weight particle’s excess weight throughout nearby regions of phase space. This may be accomplished through a technique known as particle *splitting*. If we define w_{high} to be a high energy-weight cutoff, and a particle with energy-weight w is found such that $w > w_{\text{high}}$, then we split the particle into j identical “daughter” particles with energy-weight w/j , where

$$j = \text{round} \left(\frac{w}{w_{\text{avg}}} \right). \quad (148)$$

It should be clear that such a technique successfully reduces the overall particle weight without affecting the conservation of energy. However, splitting tends to increase the overall calculation time of the Monte Carlo calculation. Additionally, if w_{high} is improperly chosen (for instance, if $w_{\text{high}} < w_{\text{avg}}$), the number of daughter particles could easily outstrip available computational memory. Finally, splitting only successfully distributes energy weight throughout phase space if the daughter particles have some reasonable probability of doing different things in their ensuing histories. Splitting particles that are streaming through a void region, for example, only increases runtime without reducing variance.

A technique known as *weight windows* combines Russian roulette and splitting so that any given particle’s energy-weight is restricted to reside in a possibly parameter-dependent window of interest, defined by w_{low} and w_{high} . Thus, low energy-weights *and* high energy-weights are not permitted in the problem. Since Russian roulette tends to decrease – and splitting tends to increase – the calculation time, there is no way of stating a priori whether or not weight windows will reduce the total calculation time. Assuming that the weight window is well-chosen, the net effect should be to reduce the variance in the problem solution. In the Appendix, we provide modifications to Eq. (67a) that contain source and sink terms arising from a hypothetical implementation of weight windows.

The steady state transport community has established a nearly optimal procedure for choosing weight-window centers for *local* problems – problems for which low variance is only desired in a relatively small portion of the problem (Wagner & Haghighat, 1998). Choosing a “good” weight window center for a *global* transport problem – problems for which we desire low variance everywhere – has been an area of active research. The steady-state neutron transport community has explored several means of prescribing a weight window and has appeared to converge on a methodology known as FW-CADIS (forward-weighted continuous adjoint deterministic importance sampling) (Wagner et al., 2009). In this method, two detailed deterministic calculations are performed, a forward solution followed by an adjoint solution using the inverse of the forward solution as the adjoint source, and the weight window center is then established using the adjoint solution. Cooper and Wollaber have explored using weight window centers based only on a “low-order” forward solution (Cooper, 1999; Wollaber, 2008; Wollaber & Larsen, 2009a; Wollaber & Warsa, 2013), but this procedure is known to sometimes produce particles with exceedingly long histories. Because adjoint solutions proceed backwards in time, and because, from cycle to cycle, IMC solutions are nonlinear, it is not straightforward to extend FW-CADIS to IMC. However, we are considering FW-CADIS-like methodologies (perhaps employing an approximate adjoint solution over a single time step). For now, the problem of establishing a procedure that produces a nearly optimal weight window center for IMC transport problems remains an open question.

Although we have only considered spatially dependent weight windows, it is also possible to bias the particle’s outgoing scattering direction (Zimmerman, 1997), although this may only conserve energy in expectation instead of on a per-particle basis.

6.3.4. Source Biasing

In the absence of an automated procedure for prescribing weight windows, an alternative lever arm to enhance or reduce the particle counts in particular problem regions is known as *source biasing*. Essentially, one prescribes a cell- and surface-source-dependent field throughout the problem that can weight the creation of additional particles – with correspondingly reduced particle weights – in regions that require better statistics. If one considers simulating an infinite medium, equilibrium problem with largely disparate cell volumes, then, if the particle energy-weights are roughly uniform, the smaller cell volumes will be more susceptible to stochastic noise in the particle paths and absorptions. Source biasing can therefore be used to increase the particle source populations in those regions with smaller energy-weights, reducing the noise in those regions (Long & Gentile, 2012). It is also possible to avoid emitting particles in pre-identified equilibrium regions altogether to allow those regions to stay in equilibrium without any noise (Urbatsch, 2012).

In addition to emphasizing and de-emphasizing particular problem regions as a whole via source biasing, it is also possible to bias particle sourcing in each cell to promote boundary crossings with a technique known as “fringe biasing” (Smedley-Stevenson, 2013). Obtaining low variances at cell interfaces can be important for coupling IMC to methods that require “consistency terms” at those locations (Park et al., 2013a, 2012), or it can be generally important to promote particle transport from thick, highly absorbing regions to adjoining regions.

6.3.5. Combing for Population Control

Particle *combing* is a widely used mechanism for controlling the global number of Monte Carlo particles, usually in time-dependent calculations (Booth, 1996; Gray et al., 1998; Kuilman et al., 2012). Generally speaking, one “stacks” the energy-weights of the particles in a random order; the length of the comb W is the global summation of all of the energy-weights in the particle census. The distance between the teeth of the comb is then the length divided by the desired number of new particles, M . The comb is then overlaid onto the stack of energy-weights with a random offset, and each tooth touches at least one particle. Some particles are not selected; these are killed. Other particles that are touched by N teeth of the comb are split N times. However, all particle weights are assigned the equal energy weight W/M . An example schematic of a comb is provided as Fig. 6, depicting the equally spaced comb teeth pointing from top to bottom and the stacked particle weights in a horizontal line at the bottom.

Because this procedure does not conserve energy on a per-cell basis (unless a comb is applied independently to each cell), and because this scheme requires global knowledge of the particle energy-weights in

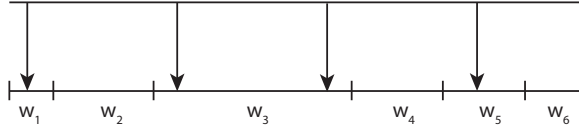


Figure 6: A recreated schematic of the particle combing procedure from (Booth, 1996). In this example, the targeted post-comb population is 4; the pre-combed population is 6, and W is the sum of the particle weights. Because the teeth of the comb (the downward arrows) do not touch particles 2, 4, or 6, these particles are killed. Because one “tooth” of the comb touches particles 1 and 5, one copy of these particles is made with weight $W/4$, and because two teeth touch particle 3, two copies of this particle are made with weights $W/4$.

each cell, it is common to use a combination of Russian roulette and splitting to achieve nearly the same result. However, the reader is cautioned that it is possible, using roulette, to kill all of the particles, and in this case it is necessary to “resurrect” a particle to ensure energy balance. It is also possible to choose the resurrected particle in such a way that it is unaffected by the parallel topology, such as by storing a value from the random number stream associated with that particle. We also note that, although particle energy may be conserved, particle momenta and frequency and angular distributions can only be conserved in expectation.

6.3.6. Momentum Tallies

IMC calculations that are coupled to hydrodynamics simulations, such as is necessary in Inertial Confinement Fusion (ICF) problems, must also account for the momentum and kinetic energy exchange between the radiation and matter. Unfortunately, momentum is a vector quantity that can take both negative and positive values,

$$\text{Momentum} = \iiint \mathbf{\Omega} \frac{\sigma}{c} I(\mathbf{x}, \mathbf{\Omega}, \nu, t) d\mathbf{\Omega} d\nu dV, \quad (149)$$

and signed tallies that depend on particle direction are notoriously high in variance. Hykes and Densmore examined several variance reduction mechanisms for this tally, including absorption, collision, and track-length estimators (Hykes & Densmore, 2009). It is also possible to reduce variance by simply not tallying momentum from isotropic emission, which is zero in expectation. Cleveland et al. have also proposed a means of tallying momentum in curvilinear coordinates for problems employing continuous energy deposition, as the particle angle is not constant throughout its flight in those geometries (Cleveland et al., 2015).

7. Conclusions

The Implicit Monte Carlo method has, despite its flaws, remained a relevant means of obtaining solutions of the thermal radiative transfer equations. As we have demonstrated, this is likely because, (1) its solutions are “robust” in the sense that they remain positive and energy-conserving for physically meaningful problem data, (2) its solutions are accurate for a reasonable range of grid spacings, (3) its Monte Carlo interpretation and implementation can draw upon many (but not all) of the storied history of Monte Carlo procedures for linear transport phenomena, easing its barrier to development, and (4) its solutions can be accelerated via the random walk and discrete diffusion methods, remedying the severest deficiencies of the approach. In this review we have re-derived the IMC equations in order to explicitly call out its approximations – both a linearization and time discretization – and, we have put the IMC equations in context with other mathematical models that employed similar approximations. In particular, we have demonstrated that the IMC equations are equivalent to the Taylor-expanded equations used as the starting point in deterministic methods for gray TRT problems. We provided a Monte Carlo interpretation of the IMC equations, using energy-weights, and emphasized the distinction between energy-weights and photon energies. We outlined a simple IMC implementation.

We considered several temporal accuracy issues relevant to IMC, including its $O(\Delta t)$ accuracy as $\Delta t \rightarrow 0$, we presented a new, linearized stability analysis for opacities with monomial temperature dependence, and we discussed how solutions of the IMC equations are subject to violations of the maximum principle. We

discussed how the IMC equations have been demonstrated to not retain the equilibrium diffusion limit due to its semi-implicit treatment of the temperature, although the equations do produce reasonable solutions in that limit. We summarized recent research that has attempted to address temporal accuracy in the IMC equations.

The imposition of a discrete temperature representation in the IMC equations also presents its own issues, in particular, “teleportation error” and potential inaccuracies in the equilibrium diffusion limit depending upon the details of the subcell spatial temperature profile, or “tilt”. The primary symptom of both of these issues is to artificially increase the numerical wavespeed of a thermal wave.

Finally, we covered a handful of algorithmic techniques that are particularly relevant to the stochastic solution of the IMC equations, including parallel computation considerations and the diffusion acceleration methods of “random walk” and DDMC. We also introduced several energy-conserving strategies for variance reduction, including absorption weighting, continuous energy deposition, weight windows, source biasing, and particle combing.

IMC, with surprisingly little modification, has had an impressively long and storied past as a method of obtaining approximate solutions of the equations of thermal radiative transfer. It is this author’s opinion that it will continue to be relevant for the foreseeable future, but, as nonlinear solver technology continues to evolve, its primary deficiency – that it is not truly time-implicit – will be its eventual downfall, both from the standpoint of accuracy (maximum principle violations in particular) and efficiency (effective scattering being an unnecessary expense in nonlinear methods). We speculate that, even under this scenario, the Fleck & Cummings IMC linearization will remain pertinent as a preconditioner step in the context of a nonlinear method, or perhaps be relegated to problems whose nonlinear coupling is relatively weak. Either way, our community is indebted to Fleck & Cummings for having the insight to translate a nonlinearly coupled system of equations into a lower triangular system containing a linear transport problem amenable to a Monte Carlo interpretation.

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Appendix

In this Appendix, we discuss modifications to Eq. (67a) that account for source and sink terms due to a hypothetical weight window treatment to further illustrate how variance reduction mechanisms can be represented in a Monte Carlo transport equation. In this treatment, we suppose that a space-dependent weight window center (or average) is provided as $A(\mathbf{x})$ for $\mathbf{x} \in V$, where $A(\mathbf{x}) \in C^1$, and $0 < A(\mathbf{x}) < \infty$. We also are given a $k \in \mathbb{N}$ so that $U(\mathbf{x}) = kA(\mathbf{x})$ is the maximum, *upper* weight in the weight window and $L(\mathbf{x}) = A(\mathbf{x})/k$ is the minimum, *lower* weight in the weight window. In other words, k is the number of times that a particle will split its weight from $w = U$ to create k particles at weight $w = A$, and k is the multiplicative weight increase to bring particles from $w = L$ to $w = A$ in a Russian roulette process (assuming that $k - 1$ particles are killed). The energy-weights are therefore constrained by “boundary conditions” such that $L(\mathbf{x}) \leq w \leq U(\mathbf{x})$.

For simplicity of presentation, we consider Monte Carlo transport processes in which the weight w is constant (no continuous energy deposition or absorption weighting, for example). Therefore, particles stream freely until they undergo collisions or until they intersect with the isosurfaces described by $w - L(\mathbf{x}) = 0$, at which point they undergo Russian roulette, or $w - U(\mathbf{x}) = 0$, at which point they undergo splitting.

We derive the removal term due to particle splitting by considering the net number of Monte Carlo particles removed along a path parameterized by moving from s to $s + \Delta s$ (along the path $\mathbf{x} = \mathbf{x}_0 + \boldsymbol{\Omega}s'$). We presume that we are considering a path along which dU/ds is decreasing so that splitting is possible (if $dU/ds > 0$, then particles cannot intersect the surface described by $w - U(s) = 0$) and define $\Delta U = U(s) - U(s + \Delta s)$. Recall from the definition in Eq. (61) that $M(s, w)$ is the number of Monte Carlo particles at s and w per volume, per solid angle, and per energy-weight. Then

$$\overline{M}(s) \equiv \int_{L(s)}^{U(s)} M(s, w) dw = \text{the number of Monte Carlo particles at } s \text{ per volume and solid angle.}$$

Recognizing that all of the particles with weights between $U(s)$ and $U(s + \Delta s)$ will undergo splitting as they stream a distance Δs , we calculate the number of particles removed due to splitting to be

$$\overline{M}(s + \Delta s) = \overline{M}(s) - \int_{U(s)}^{U(s + \Delta s)} M(s, w) dw. \quad (150)$$

Because Δs is assumed small, we may write

$$\begin{aligned} \overline{M}(s + \Delta s) - \overline{M}(s) &= -\Delta U M(s, w = U(s)) + O(\Delta s^2), \\ \frac{\overline{M}(s + \Delta s) - \overline{M}(s)}{\Delta s} &= -\frac{\Delta U}{\Delta s} M(s, w = U(s)) + O(\Delta s), \end{aligned} \quad (151)$$

and, allowing $\Delta s \rightarrow 0$,

$$\frac{d\overline{M}}{ds} = \frac{dU}{ds} M(s, w = U(s)). \quad (152)$$

We therefore identify the net removal rate of particles due to splitting as $-(dU/ds)M(s, w = U(s))$. Recognizing d/ds as the directional derivative $\mathbf{\Omega} \cdot \nabla$ along the particle flight path, the removal term for particle splitting must have magnitude $-\mathbf{\Omega} \cdot \nabla U(\mathbf{x})$ and only occur for weights $w = U(\mathbf{x})$, that is,

$$\text{Splitting removal term} = -\delta(U - w) \mathbf{\Omega} \cdot \nabla U \tilde{I}(\mathbf{x}, \mathbf{\Omega}, w). \quad (153)$$

Another way of deriving this without appealing to number density arguments is to consider an “infinite” splitting probability per unit distance at points $s \in \mathcal{S}_{w,U}$, where $\mathcal{S}_{w,U}$ is defined to be the set of points for which $w - U(s) = 0$ as $\delta(s - \mathcal{S}_{w,U})$, and to relate the spatial Dirac delta function to the weight-dependent Dirac delta function by

$$\delta(s - \mathcal{S}_{w,U}) \tilde{I}(s) = \frac{dU}{ds} \delta(U - w) \tilde{I}(s) = -\mathbf{\Omega} \cdot \nabla U \delta(U - w) \tilde{I}(\mathbf{x}, \mathbf{\Omega}). \quad (154)$$

With the removal term so defined, the net rate in the source term due to splitting must match exactly (split particles are only redistributed in energy-weight; their positions and directions are unchanged). In the source term, particles with weights $w = U(\mathbf{x})$ split into k particles with “average” weight $A(\mathbf{x})$ (recall that $U = kA$), therefore,

$$\text{Splitting source term} = -\delta(A - w) \mathbf{\Omega} \cdot \nabla U \tilde{I}(\mathbf{x}, \mathbf{\Omega}, U) = -\delta(A - w) \mathbf{\Omega} \cdot \nabla U kA cM(w = U). \quad (155)$$

We have used the definition of $\tilde{I}$ to call attention to the detail that $\tilde{I}(U)$ may be interpreted as kM particles of weight A or as M particles of weight kA . Incorporating Eqs. (153) and (155) into Eq. (67a), assuming that $Q_m = Q_r = 0$ to simplify presentation, we have,

$$\begin{aligned} \frac{1}{c} \frac{\partial \tilde{I}}{\partial t} + \mathbf{\Omega} \cdot \nabla \tilde{I} + \sigma_n \tilde{I} - \delta(U - w) \mathbf{\Omega} \cdot \nabla U \tilde{I}(\mathbf{x}, \mathbf{\Omega}, w) &= \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \iiint \delta(w - w') (1 - f_n) \sigma_n \tilde{I} d\Omega' d\nu' dw' \\ &+ f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{cU_{r,n}}{4\pi} \delta(w - w_0) - \delta(A - w) \mathbf{\Omega} \cdot \nabla U \tilde{I}(\mathbf{x}, \mathbf{\Omega}, U). \end{aligned} \quad (156)$$

Integrating Eq. (156) using $\int_0^{U(\mathbf{x})} (\cdot) dw$, we recover Eq. (24a), as the splitting source and removal terms cancel:

$$\begin{aligned} \frac{1}{c} \frac{\partial I}{\partial t} + \mathbf{\Omega} \cdot \nabla I + \sigma_n I - \mathbf{\Omega} \cdot \nabla U I(\mathbf{x}, \mathbf{\Omega}, U) &= \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{1}{4\pi} \iint (1 - f_n) \sigma_n I d\Omega' d\nu' \\ &+ f_n \sigma_{p,n} \frac{\sigma_n b_n}{\sigma_{p,n}} \frac{cU_{r,n}}{4\pi} - \mathbf{\Omega} \cdot \nabla U I(\mathbf{x}, \mathbf{\Omega}, U). \end{aligned} \quad (157)$$

The source and sink terms describing Russian roulette are entirely comparable; one need only replace L with U and $-\mathbf{\Omega} \cdot \nabla U$ with $\mathbf{\Omega} \cdot \nabla L$, as rouletting only occurs along directions of *increasing* L . We intend to analyze the implications of these mathematical models for various choices of $A(\mathbf{x})$ and k in a future publication.